

Quantifying uncertainty of salt body shapes recovered from gravity data using trans-dimensional Markov chain Monte Carlo sampling

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SUMMARY

Accurate delineation of salt body shapes is critical for hydrocarbon exploration. Various imaging methods based on seismic data have been developed. Due to the density contrast between salt and sedimentary rocks, gravity data have also been used as a de-risking tool to constrain the salt body shapes. However, quantifying uncertainties of the salt body shapes recovered from gravity data remains underexplored. Our goal is to understand and quantify how different constraints affect uncertainties of the salt body shapes reconstructed from gravity data. We adopt a trans-dimensional Markov chain Monte Carlo (MCMC) approach to explore the uncertainties. To address the computational challenges with MCMC sampling, we resort to two methods: sparse geometry parametrization and randomized parallel tempering. The first uses a set of simple geometries (e.g. ellipses) to approximate the complex shapes of salt bodies, greatly reducing the number of parameters to be sampled and making the MCMC approach computationally feasible. The second serves to further improve the acceptance ratio and computational efficiency. To quantify the uncertainties of the recovered salt body shapes, we design several scenarios to simulate different constraints on the top boundary of salt bodies from seismic imaging. We develop a new method to impose structural constraints on the top boundaries of salt bodies. This new method combines a set of fixed ellipses with randomly sampled ellipses through a concave hull. The results from different scenarios are compared to understand how uncertainties are reduced when stronger constraints are imposed. In addition, to make our uncertainty quantification results more relevant for practitioners, we propose to compute the salt probability models which show the spatial distribution of probabilities of salt materials at each cell. Finally, we investigate the effect of an uncertain salt density on the salt body reconstruction and the case of depth-varying densities in the sedimentary background. We apply our methods to the modified 2-D SEG-EAGE and Sigsbee salt models and quantify the uncertainties of the recovered salt body shapes in different scenarios. Our results highlight the importance of properly interpreting the uncertainty estimates in light of prior information and information content in the data.

Key words: Numerical modelling; Probability distributions; Statistical methods; Gravity; Inversion.

1 INTRODUCTION

Inferring the salt body shapes from geophysical measurements has been an active area of study for decades due to their significant relevance to the hydrocarbon reservoir locations. Numerous salt imaging methods have been developed. Some recent works on salt body imaging use level set methods by solving a shape optimization problem in the framework of seismic full waveform inversion (e.g. Lewis *et al.* 2012; Guo & de Hoop 2013; Kadu *et al.* 2017; Datta *et al.* 2019; Dahlke *et al.* 2020). Oh *et al.* (2020) performed deep learning inversion of electromagnetic data to delineate salt bodies.

Gravity data has also been used for salt reconstruction because notable density contrasts exist between salt bodies and the surrounding rocks. Methods of salt imaging based on gravity data include discrete-valued inversion (Maag-Capriotti & Li 2021), level sets (e.g. Li *et al.* 2016; Rashidifard *et al.* 2021), binary formulation (Krahenbuhl & Li 2006), adaptive learning (Silva Dias *et al.* 2011) and simulated annealing (e.g. Nagihara & Hall 2001; Roy *et al.* 2005). However, most works on salt reconstruction using gravity data are deterministic in that these methods typically produce one ‘optimal’ salt model without quantification of uncertainties for the reconstructed salt body shape.

Quantifying uncertainty of geophysical inverse problems can be achieved in the Bayesian inference framework (e.g. Metropolis *et al.* 1953; Hastings 1970; Green 1995), which typically involves Markov chain Monte Carlo (MCMC) sampling (e.g. Metropolis *et al.* 1953; Hastings 1970) of a *a posteriori* probability density (PPD) given prior information and observed data. Green (1995) proposed trans-dimensional MCMC where random walks jump into different states characterized by different model dimensions (e.g. Agostinetti & Malinverno 2010; Blatter *et al.* 2018; Zhang *et al.* 2018). In the deterministic framework, a qualitative assessment of uncertainties can be explored by varying initial or reference models (Kelbert *et al.* 2012). Quantitative analysis of uncertainties can be achieved by analysing linearized model covariance matrix (Alumbaugh & Newman 2000). Recently, Wei & Sun (2021) take advantage of a mixed L_p norm regularization to quickly generate a large sequence of equivalent models, based on which the uncertainties of 3-D gravity inversions are quantified.

The goal of our work is to understand how different constraints on the top boundary affect uncertainties of the salt body shapes reconstructed from gravity data, assuming the density contrast between salt materials and the surrounding rocks is known. To achieve our goal, we designed multiple scenarios, each of which simulates a different constraint. For each scenario, we performed trans-dimensional MCMC (Green 1995) using gravity data.

A well-known bottleneck of Monte Carlo sampling is its prohibitive computational cost, especially for large-scale high-dimensional problems. There are several practical ways to reduce the computational costs. The first and probably the most straightforward is to reduce the number of unknowns via sparse model representations, such as, Voronoi tessellation (e.g. Bodin *et al.* 2009; Zhang *et al.* 2018; Ghalenoei *et al.* 2021), trans-dimensional trees (Hawkins & Sambridge 2015) and Gaussian processes (Blatter *et al.* 2021). The second strategy is to leverage the gradient information in the Monte Carlo sampling. Methods in this category include Langevin (Girolami & Calderhead 2011), Hamiltonian (Fichtner *et al.* 2019), trans-dimensional Hamiltonian (Sen & Biswas 2017) and stochastic-Newton (e.g. Martin *et al.* 2012; Zhao & Sen 2021; Zhao *et al.* 2022). The gradient information greatly accelerates the convergence and improves the ‘mixing’ of traces. In addition, tempering (e.g. Sambridge 2014) has been used to raise the acceptance ratio in the Monte Carlo sampling via manipulating the standard deviation (controlled by the temperature) of PPD and has seen various successful applications (e.g. Dosso *et al.* 2012, 2014; Bottero *et al.* 2016; Blatter *et al.* 2019; Deng *et al.* 2022). Finally, optimization boosted MCMC sampling framework performs optimization at each sampling step to generate a better candidate model and improve the acceptance rate. Methods in this category include randomize-then-optimize (RTO; e.g. Bardsley *et al.* 2014; Blatter *et al.* 2022a, b), randomized maximum likelihood (RML; Chen & Oliver 2012) and randomized maximum a posteriori (rMAP; Wang *et al.* 2018).

In our work, to tackle the computational challenges, we used sparse geometry parametrization and tempering. Sparse geometry parametrization allows us to represent salt body shapes through a set of simple geometries (e.g. ellipses or ellipsoids) that can be parametrized by a small number of variables (e.g. Panagiotakis & Argyros 2016; Datta *et al.* 2019). For instance, an ellipse can be characterized by five parameters, namely, coordinates of the centroid, the major and minor semi-axes as well as its orientation. Also, we implemented parallel tempering following Sambridge (2014) to further improve the acceptance ratio. Parallel tempering allows swapping models at two different temperature levels to avoid the barriers of low probability regions. If the swapping is implemented in a random manner, it is referred to as randomized parallel tempering (Sambridge 2014).

The recent work of Ghalenoei *et al.* (2021) adopt the Voronoi tessellation to recover the salt body using gravity and magnetic data sets in the trans-dimensional Bayesian joint inversion framework. The entire model is partitioned into three subspaces (or, units) corresponding to, from near surface to deep, sediment, salt and basement. Each geological unit is associated with unique ranges of density and susceptibility values, which helps constrain the salt body recovery. Ghalenoei *et al.* (2022) further compute the alpha-shape based on the nuclei of Voronoi cells or spatial nodes to improve the recovery of all three units. Our work differs from Ghalenoei *et al.* (2021, 2022) in the following aspects. First, we use ellipses to obtain a sparse parametrization of the salt body, instead of Voronoi cells. Secondly, we assume the density contrast between the salt and the sedimentary rocks is known (to either a precise value or within a small range) and focus on recovering the shape of the salt body, whereas Ghalenoei *et al.* (2021, 2022) estimate the density and susceptibility values for each Voronoi cell. In a sense, our work falls into the category of geometry inversion, while Ghalenoei *et al.* (2021, 2022) belongs to the category of physical property inversion. Finally, the focus of our work is on understanding how different constraints on the top boundary affect the uncertainties of the salt body shapes recovered from gravity data.

We should also note that geometry inversion, also referred to as the *shape-of-anomaly* method (René 1986), has been around since the early days of interpreting potential field data (e.g. Talwani *et al.* 1959; Bott 1960; Talwani & Ewing 1960; Corbato 1965). The basic strategy is to assume a known physical property contrast and to estimate the locations of vertices of a polygon (or polyhedron), or the thickness of prisms (which defines the depths to the top of a basement). Some of the more recent works include, among others, Barbosa *et al.* (1997), Silva *et al.* (2006), Uieda & Barbosa (2012), Zhou (2013), Galley *et al.* (2020) and Lu *et al.* (2021). Our work is different in that we use a rigorous Bayesian inference framework to quantify the uncertainties resulting from different scenarios.

In the remainder, we first review the theory of the trans-dimensional Bayesian inference, sparse geometry parametrization and randomized parallel tempering. We then apply our methods to the SEG-EAGE salt model. We simulate multiple scenarios corresponding to different amounts of constraints typically encountered in practice. We also investigate the effect of an uncertain salt density on the salt body reconstruction, and consider the variation of density values with depths in the sedimentary background. Finally, we apply our workflow to the Sigsbee salt model characterized by a much more complex geometry.

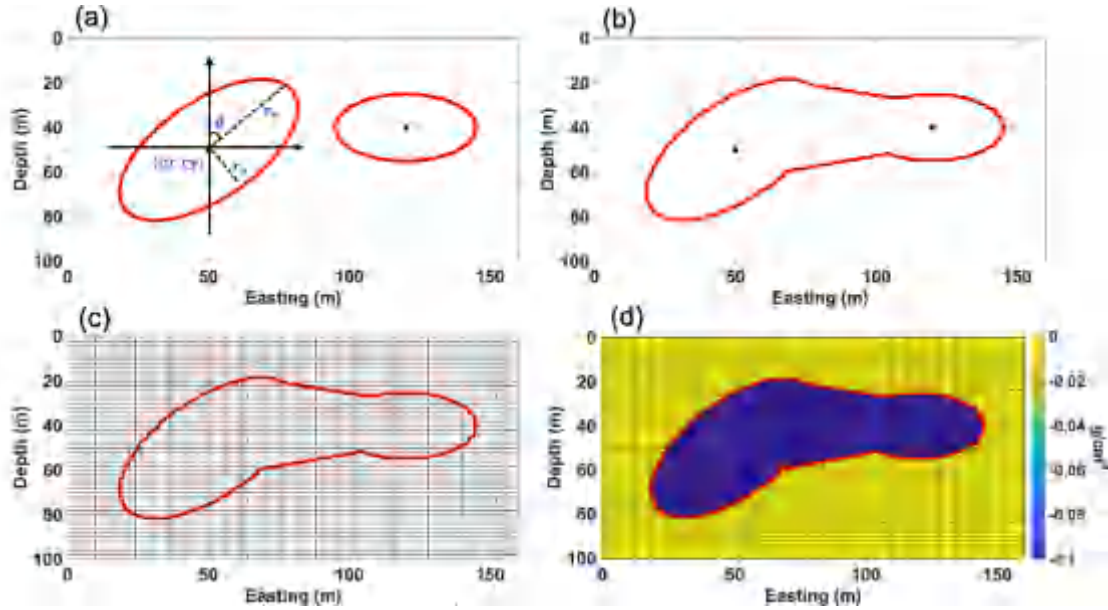


Figure 1. (a) Two randomly generated ellipses, each of which is controlled by five parameters. The boundaries of ellipses are indicated by the red lines and the centres are marked by black dots. (b) An example of the connected concave shape resulting from computing a concave hull using the two ellipses shown in (a) and an α value of 30. (c) The concave shape is discretized into a number of cells (equivalently, prisms). (d) The density contrast value of -0.1 g cm^{-3} is assigned to the cells located within the connected concave shape. The model shown in (d) is used for calculating the gravity response.

2 METHODOLOGY

2.1 Trans-dimensional Bayesian inference

According to the Bayes' rule, the *a posteriori* probability density (PPD) of trans-dimensional MCMC (Green 1995) can be written as:

$$P(\mathbf{m}, k | \mathbf{d}^{\text{obs}}) \propto L(\mathbf{d}^{\text{obs}} | \mathbf{m}, k) P(\mathbf{m} | k) P(k), \quad (1)$$

where $P()$ is a probability density function. $\mathbf{m}$, k and $\mathbf{d}^{\text{obs}}$, respectively, indicate the model parameters, number of unknowns and observed data. $P(\mathbf{m}, k | \mathbf{d}^{\text{obs}})$ is the target PPD representing the probability distribution of model parameters and dimensions given the observed data. The joint *a priori* probability density function is summarized by $P(\mathbf{m} | k) P(k)$, which contains prior knowledge or assumption about the model parameters as well as the dimension (equivalently, the number of unknowns). $L(\mathbf{d}^{\text{obs}} | \mathbf{m}, k)$ is the likelihood function, quantitatively measuring the goodness-of-fit of the observed data given a particular model $\mathbf{m}$ and the number of unknowns k .

Following Green (1995), the acceptance ratio in trans-dimensional MCMC is defined as:

$$\alpha = \min \left(1, \frac{P(\mathbf{m}^i, k^i | \mathbf{d}^{\text{obs}})}{P(\mathbf{m}^j, k^j | \mathbf{d}^{\text{obs}})} \times \frac{Q(\mathbf{m}^j, k^j)}{Q(\mathbf{m}^i, k^i)} \times |\mathbf{J}| \right), \quad (2)$$

where i and j indicate the candidate and current models, respectively. P and Q represent the posterior and proposal probabilities. In the birth-death strategy, $|\mathbf{J}| = 1$ (e.g. Agostinetti & Malinverno 2010; Sen & Biswas 2017). In our work, the model parameters are drawn from prior probability distributions, so the ratio of proposal and priori can be cancelled (e.g. Dosso *et al.* 2014; Sen & Biswas 2017). The acceptance ratio can thus be further simplified as follows:

$$\alpha = \min \left(1, \frac{L(\mathbf{d}^{\text{obs}} | \mathbf{m}^i, k^i)}{L(\mathbf{d}^{\text{obs}} | \mathbf{m}^j, k^j)} \right). \quad (3)$$

2.2 Sparse geometry parametrization

To reduce the number of parameters to be sampled in trans-dimensional MCMC, we perform sparse geometry parametrization, which allows approximating the shape of a salt body through a number of simple geometries. Each geometry can be parametrized by a small number of variables. For example, an ellipse is characterized by only five parameters, namely, coordinates of the centroid (c_x , c_y), the major and minor semi-axes (r_a , r_b), as well as its orientation θ (Fig. 1a). Fig. 2 shows the result of approximating a synthetic true salt body (in a) using five ellipses (in b). We note that other geometries are also applicable. In Fig. 2(c), we show the result of approximating the true salt body using six rectangles. Although the approximated salt body shapes resulting from the two geometries are slightly different, they are both adequate approximations of the true salt body. We have also tested other geometries such as equilateral triangles in our work and found that the choice of the basic geometries does not affect the results much. In the remainder, we focus on ellipses.

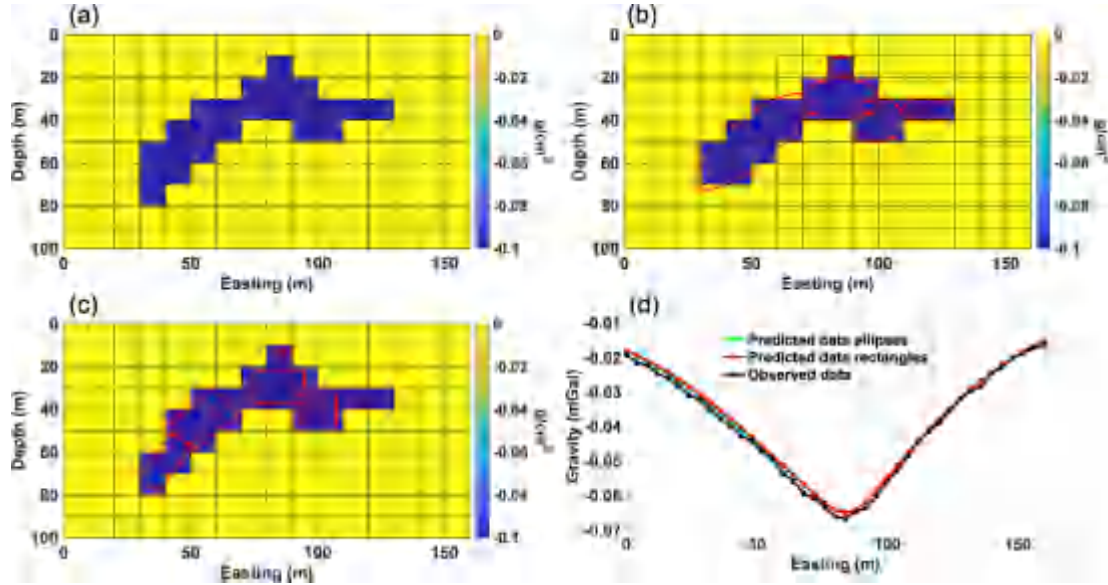


Figure 2. Simulated salt bodies using different geometries. (a) The true salt body. Panels (b) and (c) are the simulated salt bodies using ellipses and rectangles, respectively. (d) The observed data (black) for the true model and the predicted data for the simulated models using ellipses (green) and rectangles (red).

Given a total of k ellipses, the prior distribution for the x -coordinate of the centroid for the i th ellipse is assumed to be uniform, which is defined as:

$$P(c_{xi}|k) = \begin{cases} \frac{1}{c_{x\max} - c_{x\min}} & \text{if } c_{x\min} < c_{xi} < c_{x\max} \\ 0 & \text{otherwise} \end{cases}, \quad (4)$$

where $c_{x\min}$ and $c_{x\max}$ are lower and upper bounds for the x -coordinate. Since centroid coordinates are assumed to be independent among different ellipses, we obtain the joint prior distribution of the x -coordinates for all the centroids as follows:

$$P(\mathbf{c}_x|k) = \prod_{i=1}^k P(c_{xi}|k). \quad (5)$$

We can define the prior distributions for all the other parameters in much the same way. The joint prior probability density for all ellipse parametric variables is:

$$P(\mathbf{m}|k) = P(\mathbf{c}_x|k)P(\mathbf{c}_y|k)P(\mathbf{r}_a|k)P(\mathbf{r}_b|k)P(\theta|k), \quad (6)$$

where $\mathbf{m} = [\mathbf{c}_x, \mathbf{c}_y, \mathbf{r}_a, \mathbf{r}_b, \theta]$, corresponding to coordinates of the centroid, the major and minor semi-axes as well as orientation.

In our implementation of trans-dimensional MCMC, five perturbations are allowed, namely, adding an ellipse (birth), removing an ellipse (death), moving ellipses (i.e. changing the location of the centre), varying the major and minor semi-axes and adjusting the orientations of ellipses. We assign an equal probability, that is 20 per cent, to each perturbation.

Because the randomly sampled ellipses are likely to be disconnected from each other in the spatial domain (e.g. Fig. 1a), resulting in jagged geometries, we impose a smooth operator to regularize the salt body shapes. The smooth operator is implemented by computing a concave hull (Edelsbrunner *et al.* 1983) of the sampled ellipses. Fig. 1(b) shows one example of such a concave hull resulting from the two ellipses in Fig. 1(a). The continuity and convexity of a hull is determined by a tuning parameter denoted as α hereafter. In Fig. 3, we show three different concave hulls resulting from the same data points (i.e. black dots) but different α values. In Fig. 3(a), a small α value results in several disconnected triangular shapes. It is also worth noting that not all the data points (Fig. 3a) are included into the computed shapes when a small α value is used. When we increase the α values, the area and the convexity of the resulting shapes increase, as shown by Figs 3(b) and (c). A large α value tends to straighten all the most external points (Fig. 3c). In our work, we randomly sample the α values from a wide range. A randomly sampled α value is accepted or rejected according to the acceptance ratio in eq. (3). In other words, the determination of the appropriate α values is driven by the geophysical data $\mathbf{d}^{\text{obs}}$ in our work.

To calculate the acceptance ratio defined in eq. (3), we need to simulate the gravity response of the salt body in Fig. 1(b). We first discretize the entire model space into many small cells. The discretized area within the concave hull (i.e. the red curve in Fig. 1(c) defines the spatial extents of a salt body. We assume that the density contrast between the salt and its sedimentary surrounding is known, and is assigned to the cells located within the concave hull. The resulting density model in Fig. 1(d) is then used to calculate the gravity response. The predicted gravity data in Fig. 2(d) were calculated this way.

We note that, the cell based parametrization described above is only used for calculating the gravity response of a salt body. During the MCMC sampling, we are only concerned with the five parameters that define an ellipse. Assuming that there are N ellipses sampled at the

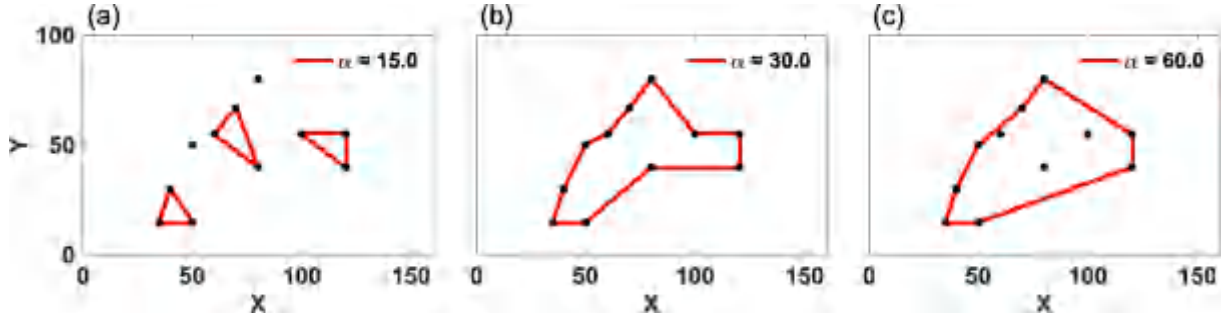


Figure 3. The resulting concave hulls (red lines) computed based upon a finite number of points (black dots) using different values of α .

current step, then the MCMC sampling would take place in a $5 \times N$ dimensional space, instead of M dimensional space, where M is the number of cells illustrated in Fig. 1(d). The use of sparse geometry parametrization results in $5 \times N < M$, making the MCMC sampling much more computationally tractable.

To summarize, the sparse geometry parametrization provides a solution to the computational challenges typically encountered in high-dimensional MCMC sampling, while at the same time, permitting adequate approximation of the complex shape of a salt body. We mention that using simple geometries such as ellipses for salt body imaging has been reported in Datta *et al.* (2019).

2.3 Randomized parallel tempering

Tempering is an effective way of raising the acceptance ratio in the Monte Carlo sampling. The tempered posterior probability distribution can be formalized as (Sambridge 2014):

$$P(\mathbf{m}|\mathbf{d}^{\text{obs}})^{1/T} \propto L(\mathbf{d}^{\text{obs}}|\mathbf{m})^{1/T} P(\mathbf{m}), \quad (7)$$

where T indicates a temperature level and a large T value increases the acceptance ratio. The idea behind the tempering is to improve the chance (equivalently, probability) of sampling in the low probability model space. The left-hand side of eq. (7) shows that the T value controls the flatness of PPD because T affects the posterior standard deviation. A high temperature level corresponding to a large standard deviation flattens the PPD, and it thus increases the probability for the originally low probability regions, allowing the sampling process to jump out of local minima. The right hand side shows that a large T value increases the variance of the observed data and relaxes the data-fitting. For a more detailed explanation of tempering, we refer the readers to Sambridge (2014).

The parallel tempering allows swapping models (i.e. $\mathbf{m}^i$ and $\mathbf{m}^j$) randomly sampled at two different temperature levels (i.e. T_i and T_j). Following Sambridge (2014), we express this exchange as:

$$(\mathbf{m}^i, T_i), (\mathbf{m}^j, T_j) \rightarrow (\mathbf{m}^i, T_j), (\mathbf{m}^j, T_i). \quad (8)$$

If the swapping is implemented in a random manner, it is referred to as randomized parallel tempering (Sambridge 2014). Fig. 4 displays our workflow of randomized parallel tempering. The minimum temperature level is commonly 1. The maximum temperature value, hereafter referred to as T_{max} , varies from one problem to another. Typically, a few experimental tests are performed to determine a proper value for T_{max} to achieve a reasonable acceptance rate, for example, 20 per cent (Earl & Deem 2005). A generally used strategy to determine a temperature ladder (i.e. a set of temperature levels) for the chains is to decrease the number of chains with logarithmic increase of temperatures, for example 16, 8, 4 and 2 chains at temperatures of $T = 1, 2, 4$ and 8 (Dosso *et al.* 2012). An alternative method is to have a proportion of chains (e.g. 25 per cent) at $T = 1$ with the rest randomly sampled from a log-uniform distribution (Sambridge 2014).

In our work, the acceptance rule (also termed as acceptance ratio) for the model exchange in the randomized parallel tempered trans-dimensional MCMC is defined as:

$$\beta = \min \left(1, \left[\frac{L(\mathbf{d}^{\text{obs}}|\mathbf{m}^i, k^i)}{L(\mathbf{d}^{\text{obs}}|\mathbf{m}^j, k^j)} \right]^{1/T_j} \times \left[\frac{L(\mathbf{d}^{\text{obs}}|\mathbf{m}^j, k^j)}{L(\mathbf{d}^{\text{obs}}|\mathbf{m}^i, k^i)} \right]^{1/T_i} \right), \quad (9)$$

where i and j are the indices of two chains tempered by different temperature levels. The use of the acceptance ratio for model exchange (eq. 9) is consistent with Metropolis–Hastings algorithm, that is if $\beta > U(0, 1)$, the models, $\mathbf{m}^i$ and $\mathbf{m}^j$, are swapped, otherwise, no model exchanges involved. More details of the swapping acceptance rule are given in the Appendix A.

As mentioned earlier, a large T value increases the acceptance ratio, but at the expense of accepting some models that fit the measurements poorly. These models would bias the mean and standard deviation of PPD. A reweighting procedure is usually performed to correct for the bias and to obtain the unbiased posterior mean ($\bar{\mathbf{m}}$) and standard deviation (var; Brooks & Frazer 2005) according to the following equations:

$$\gamma_k^j = L(\mathbf{d}^{\text{obs}}|\mathbf{m}_k^j)^{(1/T_1 - 1/T_k)}, \quad (10)$$

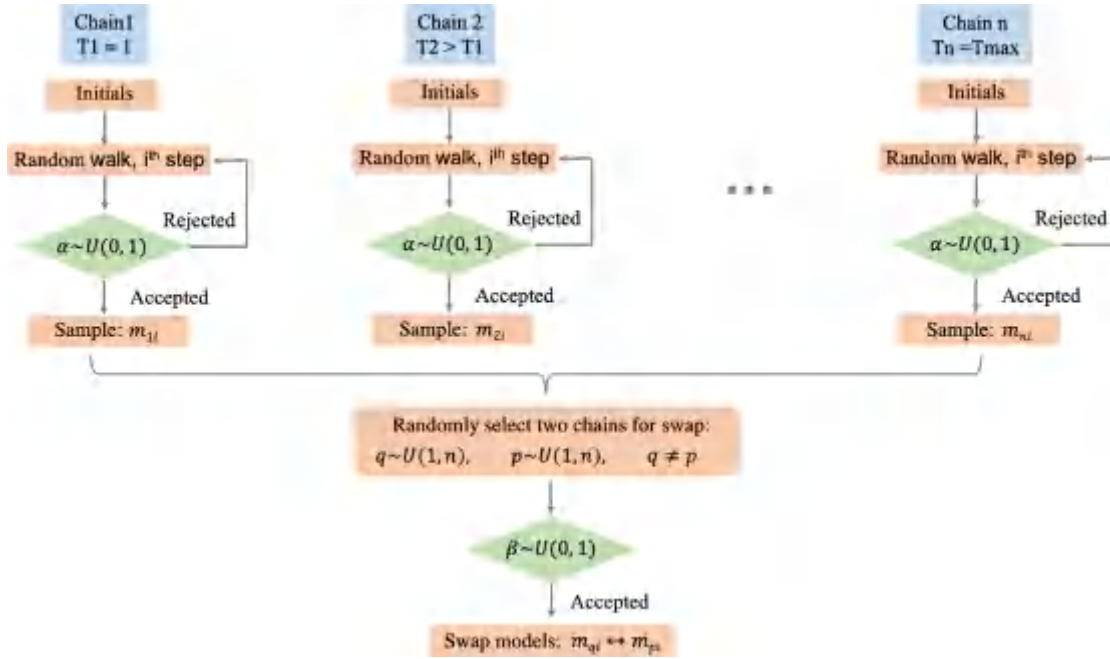


Figure 4. The flowchart of randomized parallel tempering. Each chain associates with a temperature level. The first and last chains are, respectively, characterized by the minimum and maximum temperatures, that is 1 and $T_{\max}$. The remainders are sampled from the range of $[1, T_{\max}]$. There are two levels of the *accept and reject* criteria. The first one is to accept or reject models after the steps of random walk for each chain. The other is the acceptance of swap for the two randomly selected chains out of all the chains. If swap is accepted, the current models of these two chains are exchanged.

$$\bar{\mathbf{m}} \approx \frac{1}{C} \sum_{k=1}^C \frac{\sum_{j=1}^N \gamma_k^j \mathbf{m}_k^j}{\sum_{j=1}^N \gamma_k^j}, \quad (11)$$

$$\text{var} \approx \frac{1}{C} \sum_{k=1}^C \frac{\sum_{j=1}^N \gamma_k^j (\mathbf{m}_k^j - \bar{\mathbf{m}})^2}{\sum_{j=1}^N \gamma_k^j}, \quad (12)$$

where k and j represent the indices of chains and steps, respectively. γ_k^j indicates an importance weighting parameter used to correct the bias for the j th step in the k th chain tempered by T_k . Capital C represents the total number of chains, each of which is associated with a temperature level. N is the total number of the accepted models at each chain (i.e. the last 10 per cent of models with even indices). $\bar{\mathbf{m}}$ and var represent the unbiased mean and variance using the all accepted models from all the chains. A derivation for eqs (10)–(12) can be found in Appendix B.

We point out that, in our work, the maximum *a posteriori* (MAP) model is not affected by the randomized parallel tempering. Our prior distribution is uniform, as shown by eqs (4)–(6). It then follows that the MAP model has only to do with the likelihood function and is always the one with the smallest data misfit value. Therefore, no reweighting is necessary for the MAP model.

3 SEG-EAGE SALT BODY RECONSTRUCTION

We applied trans-dimensional MCMC with randomized parallel tempering to recover a 2-D salt body extracted from the SEG-EAGE 3-D salt model. Fig. 5a shows the shape of the 2-D salt body characterized by a triangular top part, a laterally extending middle part and a dipping flank at the bottom. The density contrast between the salt body and the uniform sedimentary rocks is assumed to be -0.3 g cm^{-3} . We simulated the gravity response using the cell based approach explained earlier. We also added independent Gaussian noise with a mean of 0 mGal and a standard deviation of 0.1 mGal. We designed multiple scenarios that correspond to different constraints available in practice. In each scenario, we had 200 chains associated with multiple temperature levels and each chain contains 10 000 steps. We fixed 35 per cent of chains (i.e. 70 chains) at $T = 1$ and uniformly sampled the temperature levels from $(1, 1500]$ for the remaining chains. Samples were allowed to swap among 200 chains at each step. For each scenario, the sampling process was completed within two hours using parallelization (i.e. *parfor* in MATLAB). For uncertainty analysis and salt probability calculation, we only kept the last 10 per cent samples with even indices for each chain to increase samples' independence.

We randomly sampled a set of ellipses to approximate the 2-D SEG-EAGE salt model. Specifically, the number of ellipses were perturbed within $[2, 14]$ following the *birth and death* strategy; the orientations were uniformly drawn from $[-\pi/2, \pi/2]$, and the lengths of the semi-major and semi-minor axes were sampled from $[200 \text{ m}, 1500 \text{ m}]$ and $[200 \text{ m}, 800 \text{ m}]$, respectively. The centroid coordinates vary within a feasible region that is unique in each scenario. Constraint about the shape of the salt body is incorporated into the sampling process via a

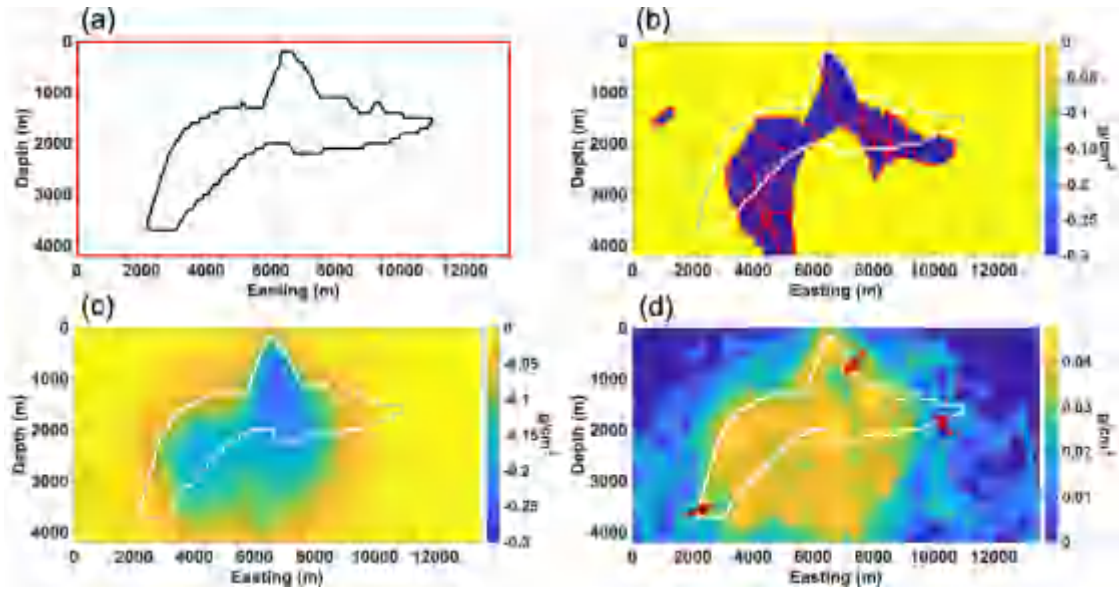


Figure 5. (a) The true salt body (black) and the feasible region (red) for the possible locations of the randomly sampled ellipses. Panel (b) shows the maximum a posteriori probability (MAP) model and the randomly sampled ellipses (red). Panels (c) and (d), respectively, display the reweighted mean model and standard deviations. Red arrows indicate the model features characterized by lower standard deviations discussed in the text.

feasible region and, if applicable, a set of fixed ellipses in each scenario. Also, the tuning parameter, α , was randomly sampled from [100, 1000] in the concave hull computation.

3.1 No constraint on the top boundary

In the first scenario, we assume no constraint on the top boundary available. The possible locations of the ellipses are within the entire model region indicated by the feasible region (i.e. the red box) in Fig. 5(a). Fig. 5(b) displays the MAP model constituted by a set of randomly sampled ellipses (red). The reweighted mean model in Fig. 5(c) recovers negative density materials roughly distributed in the top triangular area and also along the dipping part of the true salt body even without any constraint on the possible locations of salt. This indicates that our simulated gravity response does contain some information about the dipping structure of the salt body.

The reweighted standard deviations are displayed in Fig. 5(d). We observe an area of overall high standard deviations in the centre, shown by the orange colours, indicating a relatively high degree of uncertainties. We highlight three regions with relatively smaller standard deviations (and uncertainties) by the red arrows. The first one is located roughly in the centre of the top triangular feature. In the mean model in Fig. 5(c), this area is associated with a well-defined negative density anomaly with mean values close to the true salt density contrast value, that is -0.3 g cm^{-3} . The smaller standard deviations in this area indicates that most of the accepted models agree on the existence of salt materials.

A second area with smaller standard deviations is located at the east edge of the salt body. In the mean model, this area is erroneously dominated by sediments. The smaller standard deviations result from the fact that most of the accepted models ‘vote’ for sediments in this area, whereas it is salt in the true model.

It is worth noting that the low standard deviations are closely related to the information contained in the gravity data (i.e. information content). For the first low-uncertainty area at the triangular top, our data contains enough information for resolving this feature because it is at shallower depths. In this case, low uncertainty corresponds to high accuracy. In contrast, for the second low-uncertainty area, our data does not contain as much information as necessary for us to resolve the density feature, because it is deeper, and also it tapers toward the east. The larger depth and the shrinking volume both contribute to the insufficiency of enough information in the data. In this case, low uncertainty does not translate to high accuracy. Our work clearly shows that low uncertainties can occur in both well resolvable and poorly resolvable (or, unresolvable) areas in a model, and that features with low standard deviations and uncertainties are not necessarily true. Therefore, when interpreting the results, caution must be exercised to avoid associating low uncertainties with high accuracy. Indeed, low uncertainties simply mean that most of the accepted models agree on something which might be true (as in the case of the feature at the triangular top) or false (as in the case of the feature at the eastern edge).

A third area with a relatively low standard deviation is at the base of the dipping structure as indicated by the red arrow in the lower bottom of Fig. 5(d). Similar to the above case at the eastern edge of the salt body, this area, albeit associated with low uncertainties, is mistakenly assigned with sediments.

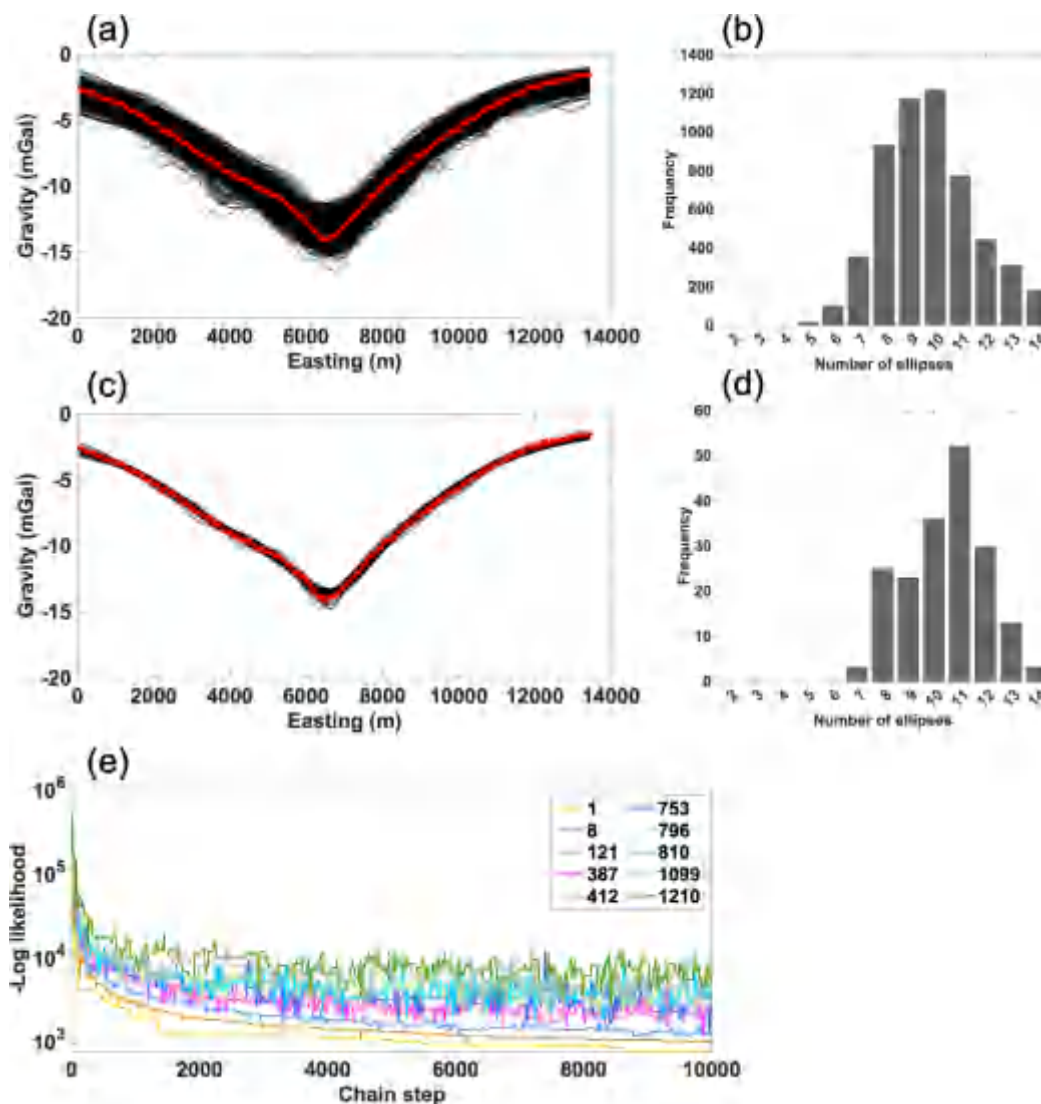


Figure 6. (a) The observed data is indicated by red. The simulated gravity data sets (black lines) correspond to the accepted models from all chains tempered by multiple temperature levels. (b) The distribution of the number of ellipses determined by the trans-dimensional MCMC method based upon the accepted models from all chains. (c) and (d), respectively, display the observed (red) and the simulated (black) gravity data as well as the number of ellipses only from the chains at the temperature level of 1. (e) exhibits the negative log likelihood of 10 chains with different temperature values that are noted in the legend and indicated by different colours. In our work, there are a total of 200 chains associated with multiple temperature levels. For the sake of brevity, only 10 chains are randomly selected and shown here.

Fig. 6(a) shows the predicted data of the accepted models from all chains at different temperature levels. Because high temperatures relax the data-fitting, we observe a large variance of predicted data indicated by the large width of black lines. The half-width is much larger than the data noise level, that is 0.1 mGal, indicating the existence of many models that are accepted but do not fit the gravity data well, hence the need of calculating the reweighted mean and standard deviation values according to eqs (11) and (12). Fig. 6(b) displays that, without any constraint, the optimal number of ellipses required to approximate the true salt body is 9 or 10, again, based on the accepted models from all chains. We also computed the predicted data (Fig. 6c) and the number of ellipses (Fig. 6d) for the accepted models at the temperature level of 1. We observe a much narrower width of the black lines, that is predicted data, indicating a much smaller variance of the predicted data (Fig. 6c). The histogram of the number of ellipses at $T = 1$ (Fig. 6d) also appears different than that from all temperatures (Fig. 6b). The peak value shifts to the right, indicating a larger number of ellipses are required to fit the gravity measurements to its noise level.

The negative logarithmic likelihood values along 10 000 steps for the different tempered chains are shown in Fig. 6(e). The chains tempered by large values (e.g. 810, 1099 and 1210) display the hairy and caterpillar features, which reflects a good mixing of traces and high acceptance ratios. At the small temperature levels (e.g. 1 and 8) we observe much flatter curves without intensive fluctuations, indicating low acceptance ratios. Although the curves from different temperatures in Fig. 6(e) differ from each other, the equilibrium states were consistently achieved after, roughly, 3000 steps.

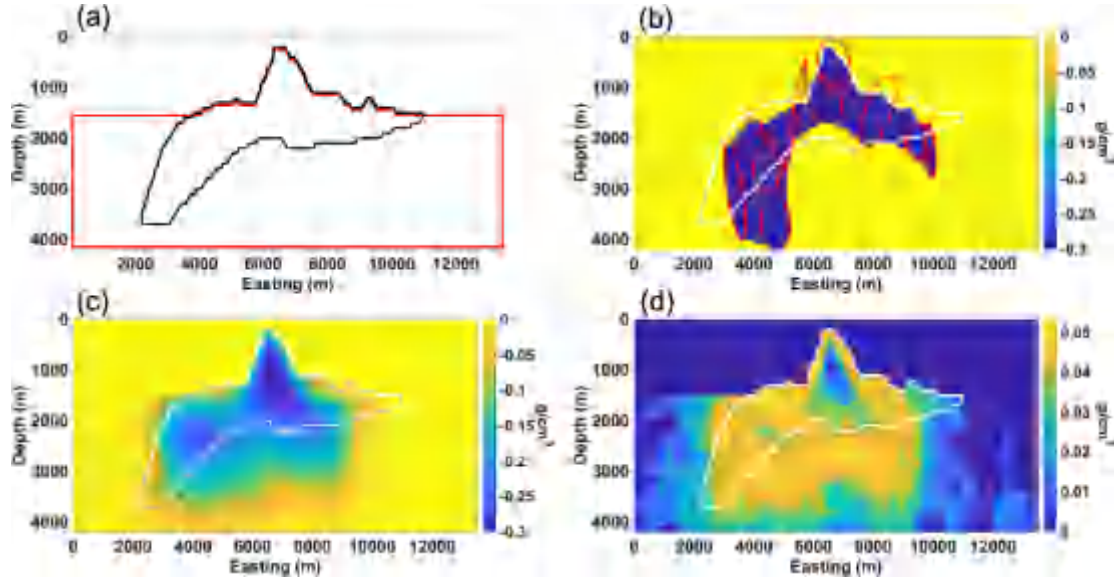


Figure 7. (a) The feasible region, bounded by red lines, is defined by a structural constraint about the shallow top of the salt body. Ellipses are randomly sampled within the feasible region. Panel (b) is the MAP model. Panels (c) and (d) are reweighted mean and standard deviations.

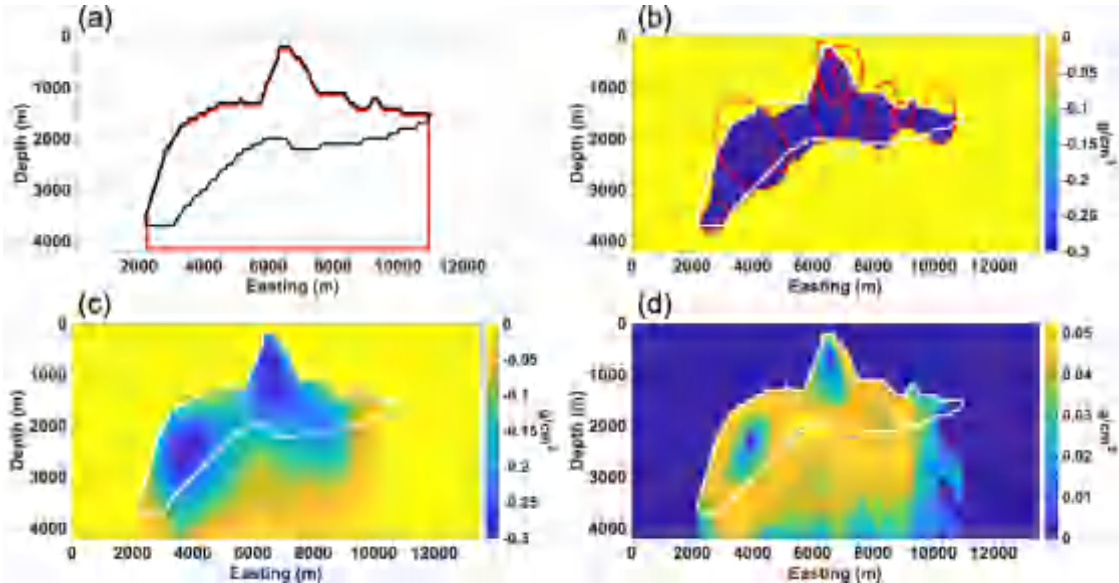


Figure 8. (a) A feasible region defined by the deep constraint about the top of the salt body. Panel (b) is the MAP model. Panels (c) and (d) correspond to reweighted mean and standard deviations, respectively.

3.2 Conservative constraint

We now assume that the shape of the top of the salt body has been imaged by seismic data. Below, we consider two cases where such constraint is used to constrain the recovery of the deeper salt body.

In the first case, we define a feasible region, bounded by the red lines in Fig. 7(a), where the randomly sampled ellipses are allowed to exist. In other words, no ellipse will be sampled outside of this region. The limited sampling region helps increase the convergence of Markov chains. The MAP model (Fig. 7b) reasonably captures the shape of the true salt body. In the reweighted mean model (Fig. 7c), we observe more salt in the dipping part. The distribution of the salt materials, represented by negative density features in blue, is in better agreement with the true salt body. Also, the top triangular structure is better recovered and shows lower standard deviations (Fig. 7d) than in previous case (Fig. 5d). The shallow constraint also helps improve the recovery of the deeper dipping part for the salt body, although it is associated with a similar level of uncertainties to the previous case (Fig. 7d).

In the second case, the geometry of the top of the salt body is assumed well known at all depths, as indicated by the more confined feasible region shown in Fig. 8(a). We will refer to this as deep constraint, which is to be contrasted with the shallow constraint described in the above paragraph. Significant improvement is observed in the MAP model shown in Fig. 8(b) where the shape of the salt body is excellently

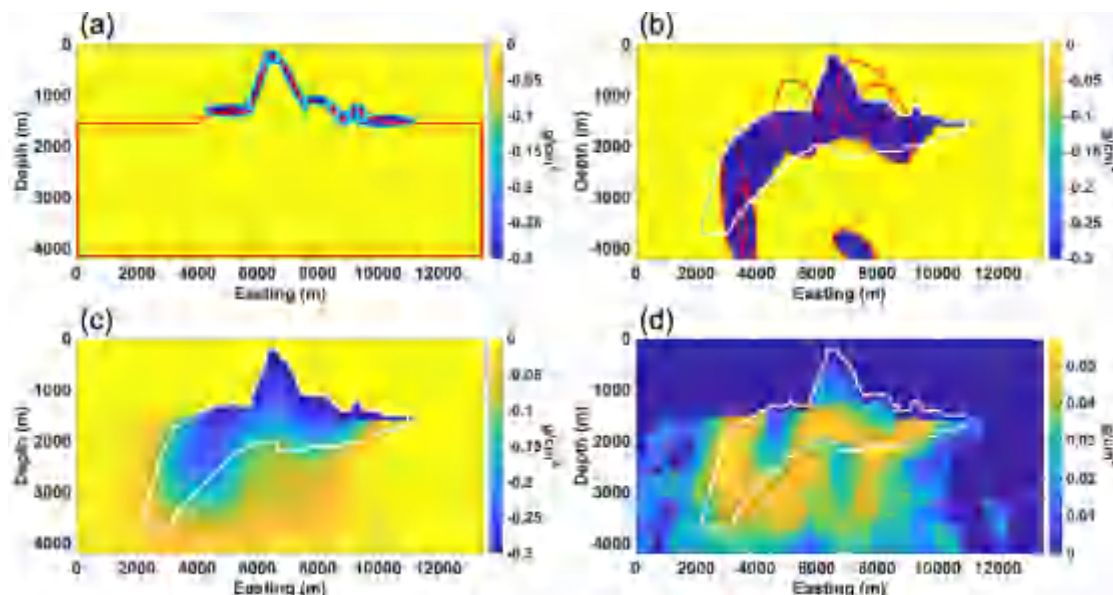


Figure 9. (a) A feasible region defined by the shallow constraint and a set of fixed prior ellipses (cyan). The true salt density contrast value is assigned within these ellipses. Panel (b) is the MAP model. Panels (c) and (d) display reweighted means and standard deviations, respectively.

recovered after imposing the deep constraint. The reweighted mean model in Fig. 8(c) also shows marked improvement. The overall shape of the reweighted mean model starts to better reflect the true shape of the salt body. We observe that more salt materials are distributed in the deeper dipping part of the salt body (Fig. 8c). Interestingly, they correspond to an area of low standard deviations in Fig. 8(d). In this case, the low standard deviations correspond to high accuracy. However, the low standard deviations at the east edge of the salt body do not, as discussed previously.

To further improve salt reconstruction, we extend the constraint introduced above, which only constrains the overall feasible region of the salt body, one step further by constraining the salt materials immediately under the top boundary. To distinguish them, we refer to the former one as conservative constraint and the latter one as informative constraint, which is detailed later.

3.3 Informative constraint

The basic idea of the informative constraint is to use several ellipses to approximate the salt materials immediately below the top boundary. The locations, sizes and orientations of these ellipses are fixed based on the constraint (e.g. from seismic imaging) throughout the MCMC sampling process. True density value for the salt body is also assigned to them. Fig. 9(a) shows these fixed ellipses (in cyan) along with a feasible region defined by the shallow constraint discussed in the preceding section. To recover the rest of the salt body, we randomly sample ellipses within the feasible region bounded by the red lines (Fig. 9a). Different from the fixed ellipses explained above, these randomly sampled ellipses are subjected to the five perturbations discussed in Section 2. Therefore, there are two types of ellipses in our work. The fixed prior ellipses provide constraints to the top of salts, whereas the randomly sampled ellipses are used to recover the main body of salts. The two types of ellipses are connected via computing a concave hull.

The MAP model (Fig. 9b), again, well reconstructs the shape of the true salt body. The reweighted mean model (Fig. 9c) shows an excellent recovery of the top part of the true salt model. We observe more salt materials distributed along the top of the salt body as well as the shallower portion of the dipping structure. Regions with density contrast values close to -0.3 g cm^{-3} (in deep blue) correspond to lower standard deviations in Fig. 9(d). These lower standard deviations are associated with higher accuracy because of the imposed informative constraint. Higher standard deviation values appear around deep boundaries (Fig. 9d), which is reasonable because only the top of the salt is constrained by fixed prior ellipses.

We also combined the deep constraint described in the preceding section with the fixed ellipses. A graphical representation is given in Fig. 10(a). We observe that the MAP model in Fig. 10(b) clearly delineates the detailed structure of the salt body. The mean density model in Fig. 10(c) also shows dramatic improvement compared with all previous scenarios. The dipping part of the salt body and the laterally extending middle part are both much better defined. The areas of high standard deviations roughly correspond to the lower boundary of the salt model (Fig. 10d). Determining the lower boundary of a salt body is usually challenging in practice. Our work shows that, if properly constrained, the standard deviation distribution, such as that in Fig. 10(d), might provide some useful information for the lower boundary.

3.4 Salt probability models

Visualizing and comparing the MAP models, the reweighted mean models and the standard deviation models help us understand how different constraints affect the recovery of the salt body shapes. It is also the most common way of analysing the accepted models from MCMC sampling

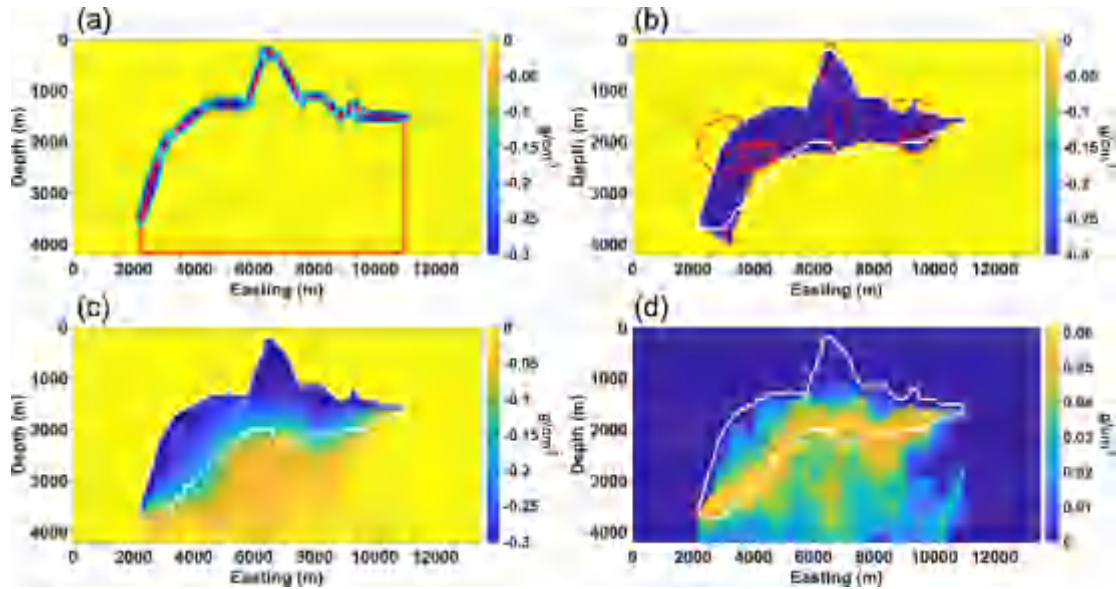


Figure 10. Panel (a) shows a feasible region defined by the deep constraint and fixed ellipses. Panels (b)–(d) are the corresponding MAP, reweighted means and standard deviations.

in the geophysics literature. However, it does not answer the question: given any location in the model region, what is the probability of the existence of salt at that location? From a practical point of view, this is the question that geophysics practitioners are most concerned about. This question is also related to another one: what is the probability of a drill encountering salt at a depth of, say, 2000 m?

To answer these two questions, we computed the salt probability at each location. First, for each accepted model at $T = 1$, we reparametrized it using the cell-based approach illustrated in Fig. 1. We then converted the resulting cell-based models into binary models where 0s and 1s represent, respectively, the sedimentary background and the salt body. The accepted models at higher temperatures were not used here because most of them do not fit the gravity measurements well. In the next step, we simply counted the frequency of 1s occurring at each cell location and normalized it by the total number of binary models obtained from the previous step. We interpret the normalized frequency values as an approximation of the salt probability at each location.

Fig. 11(a) displays the salt probability model when no seismic constraint was used. The top triangular area is associated with high salt probabilities of 80 per cent and above. In addition, we observe another area with salt probabilities of 60–70 per cent located roughly in the centre of the dipping structure (within the depth range of 2000–3000 m). Considering that no constraint from seismic imaging is used in this case, we find the result to be encouraging. The salt probability model in Fig. 11b corresponds to the shallow conservative constraint (Fig. 7a). We notice that not only is the area of higher probabilities extended, but also the probability values are elevated. For example, the core of the top triangular structure is now dominated by a probability of 100 per cent. In Fig. 11(c) when deep conservative seismic constraint is applied, the overall shape of the salt body, including the dipping structure, is well recovered. Figs 11(d) and (e) show, respectively, the salt probability models when shallow and deep informative constraint is used. Not surprisingly, the ‘best’ salt probability model is achieved with the deep informative seismic constraint.

Fig. 11 clearly shows how salt probabilities change when different constraints are imposed. We believe that it also helps answer the two questions raised above. We point out that, throughout our work, no depth weighting or its variants (e.g. distance weighting or kernel weighting) is applied.

3.5 Uncertain salt density values

In the proceeding sections, we randomly sampled a set of ellipses, computed a concave hull, and assigned the true salt density contrast, that is -0.3 g cm^{-3} , to the materials within the concave hull shape. We made an implicit assumption that the correct salt density or density contrast value is known accurately. Unfortunately, in practice, it is more often that a range for the salt density (or, density contrast) values rather than a single value is available. In this scenario, instead of using a single constant density contrast value (i.e. -0.3 g cm^{-3}), we consider two cases where the density contrast values are uncertain. In the first case, we consider a narrower range $[-0.35 \text{ g cm}^{-3}, -0.25 \text{ g cm}^{-3}]$. A wider range $[-0.4 \text{ g cm}^{-3}, -0.2 \text{ g cm}^{-3}]$ is considered in the second case. These two ranges correspond to two different levels of uncertainties for the salt density values. Both of these two cases are constrained by the deep informative constraint (Fig. 10a). In the forward modelling, a value of density contrast is randomly sample from the respective ranges and assigned to the concave hull.

The MAP model for the first case is shown in Fig. 12(a). The shape of the salt body is satisfactorily recovered including the deepest dipping structure. The reweighted mean and standard deviations are generally consistent with the models recovered by using the true density

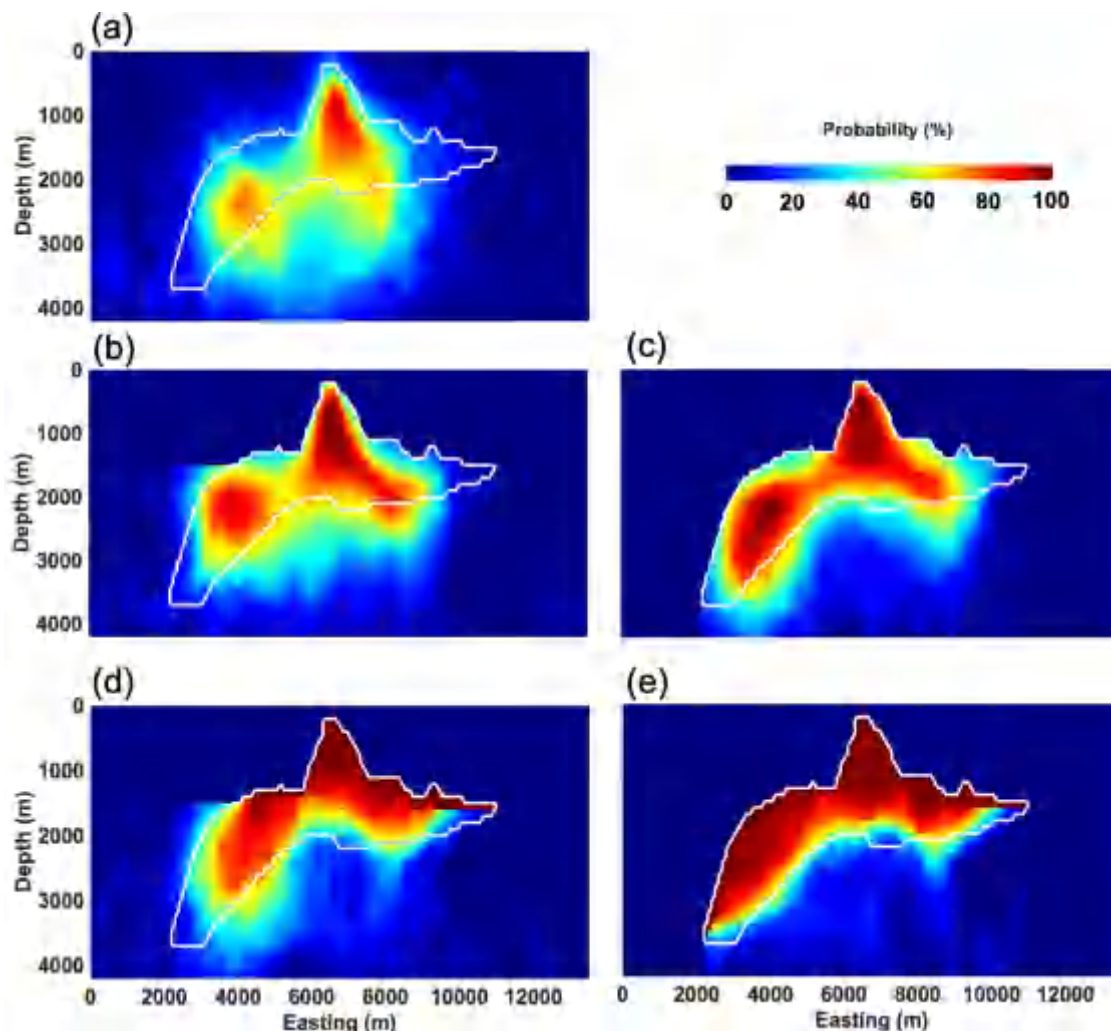


Figure 11. Salt probabilities computed based upon the accepted models associated with the temperature level of 1. Panels (a) to (e) display the salt probability models corresponding to, respectively, no constraint on the top boundary (Fig. 5a), shallow (Fig. 7a) and deep (Fig. 8a) conservative constraint as well as the shallow (Fig. 9a) and deep (Fig. 10a) informative constraint.

contrast value directly (Figs 10c and d). The bottom part of the salt model shows higher uncertainties reflected by the wider yellow area in Fig. 12(c) compared with the Fig. 10(d).

Figs 12(d)–(f) display the recovered models with the randomly sampled salt density contrast values ranging from $[-0.4 \text{ g cm}^{-3}, -0.2 \text{ g cm}^{-3}]$. The shape of the MAP model (Fig. 12d) is reasonably recovered, with extra salt materials shown at three locations below the bottom boundary of the salt body. The maximum estimated density contrast value is -0.26 g cm^{-3} which is lower than the true value -0.3 g cm^{-3} . We note that the colour of salt model in Fig. 12(d) is lighter than Fig. 10(b) using the same colourbar. The reweighted mean model (Fig. 12e) successfully recovers the top features of the salt body, but with a poorly defined bottom structure. We observe extra salt materials below the bottom boundary of the true model (delineated by the white line shown in Fig. 12e). The reweighted standard deviations (Fig. 12f) display high uncertainties in a much larger area extending to the depth of, roughly, 4000 m, compared with Figs 10(d) and 12(c).

This experiment clearly shows that the uncertainty of the salt density values does affect the recovery of the shape of the salt body as well as its uncertainties. The smaller uncertainty for the salt density values results in better recovery of the salt body with less uncertainties, vice versa. This is not a new finding. But we believe that our work is the first to rigorously quantify and compare the uncertainties of the salt body reconstruction when different salt body densities are considered. The rigour stems from the use of MCMC-based Bayesian inferences.

3.6 Density contrast variation with depth

So far, we have assumed a constant density for the sedimentary background. In this section, we consider the density variation with depth in the sedimentary rocks.

One of the significant challenges in salt body imaging using gravity data is the lack of the gravitational signal response at the nil zone where the salt density is identical to the density of the background sediments. Following Hudec *et al.* (2009), the true salt density is set to

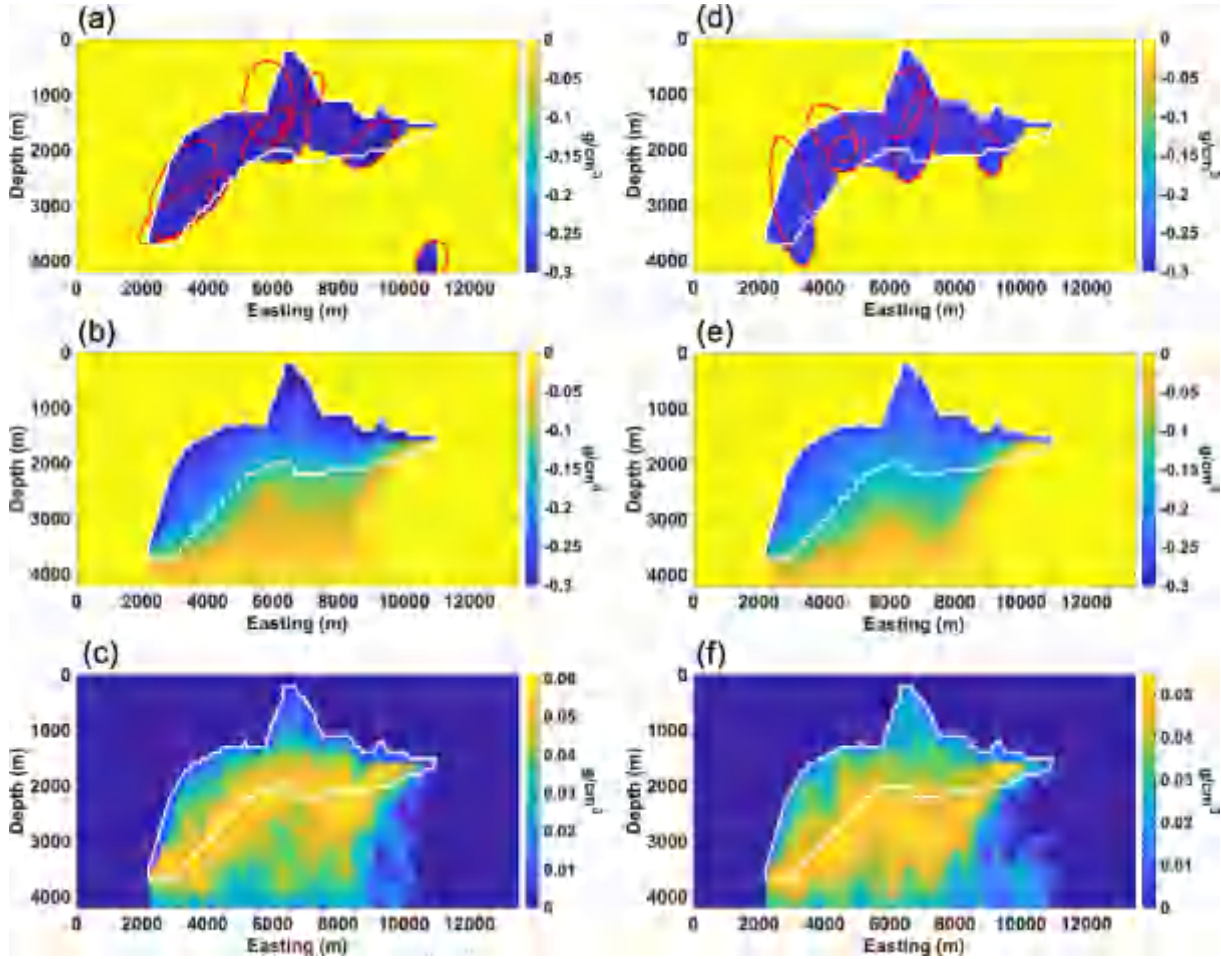


Figure 12. Panels (a)–(c) represent the MAP model, reweighted mean and standard deviations using the deep informative constraint shown in Fig. 10(a). In this case, the salt density contrast value is randomly sampled from $[-0.35, -0.25] \text{ g cm}^{-3}$. Panels (d)–(f) correspond to the recovered MAP model, reweighted mean and standard deviations with the randomly sampled salt density contrast ranging from $[-0.4, -0.2] \text{ g cm}^{-3}$. The true density contrast of the salt body is -0.3 g cm^{-3} .

2.2 g cm^{-3} and the background sedimentary density versus depth follows the equation of $\rho = 1400 + 172z^{0.21}$, where z indicates the depth below the mudline. We obtain the density contrast variation by subtracting the background density from the salt density (Fig. 13a). The 0 g cm^{-3} density contrast shown in Fig. 13(a) corresponds to the depth of nil zone that is, approximately, 1600 m. We then extended the density contrast variation curve to the entire 2-D model space shown in Fig. 13(b). Following Maag-Capriotti & Li (2021), we performed element-wise multiplication of the background density contrast model (Fig. 13b) with the binary salt model (Fig. 13c) and obtained the true salt body with density contrast variation (Fig. 13d). Based on the salt model in Fig. 13(d), we simulated the observed gravity data and added independent Gaussian noise with the standard deviation of 0.02 mGal (Fig. 13e).

We reconstruct the salt model (Fig. 13d) in two cases: no constraint on the top boundary and deep informative constraint. In the first case, the possible locations of the randomly sampled ellipses are within the entire model region indicated by the red box in Fig. 5(a). The MAP model (Fig. 14a) captures some of the main features of the salt body, such as the top triangular structure and part of dipping flank feature, but significant inconsistencies between the MAP model and the true salt body are also obvious. The reweighted mean model (Fig. 14b) shows positive anomalous features distributed at the top triangular area corresponding to the relatively lower standard deviations (Fig. 14c). We also see negative materials located at the deep dipping structures in the reweighted mean model (Fig. 14b). These negative features are also characterized by the lower standard deviations (Fig. 14c). We note that, in order to better visualize the standard deviations around the nil zone, we normalized the reweighted standard deviations by the absolute values of the mean model. Fig. 14c shows the normalized standard deviations, which are unitless. The same is true for Fig. 14f (and Fig. 19c, f and i).

In the next experiment, we imposed the deep informative constraint where several ellipses are fixed to approximate the top of salt and a set of additional ellipses are randomly sampled within the feasible region (Fig. 10a) to reconstruct the main salt body. The MAP model (Fig. 14d) is highly consistent with the true model (Fig. 13d). We observe much better recovery of the dipping part of the salt body (Fig. 14e). The negative-density materials are distributed in the correct position. In Fig. 14(f), the bulk of the salt body is associated with lower standard deviations, that is less uncertainty. Another notable observation is that the top boundary of high standard deviations, coded by the yellow colour in Fig. 14(f), distributes roughly along the bottom boundary of the salt body.

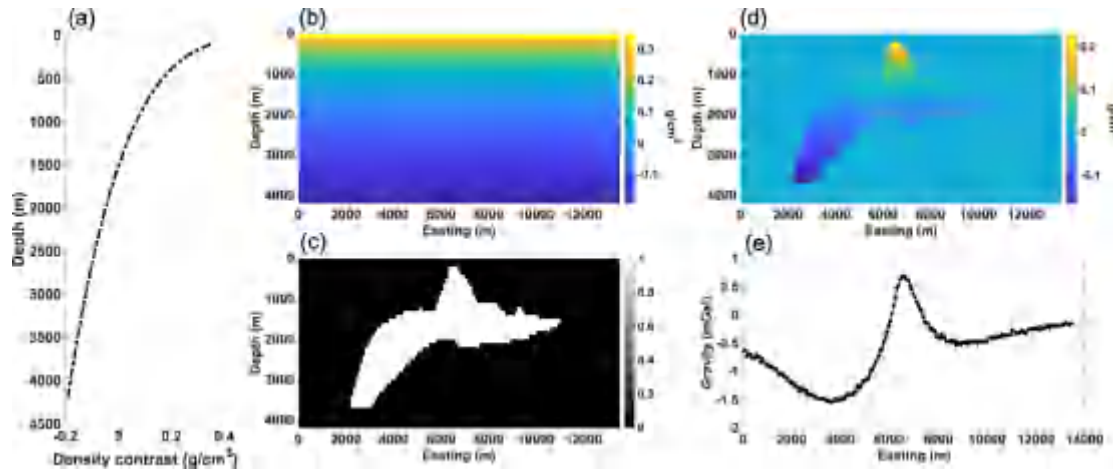


Figure 13. (a) Salt density contrast (i.e. subtracting background density from salt density) variation with depth. Following the Hudec *et al.* (2009), the salt density is set to 2.2 g cm^{-3} and the background sedimentary density versus depth follows the equation of $\rho = 1400 + 172z^{0.21}$, where z indicates the depth below the mudline. We note that the 0 density contrast value corresponds to the depth of nil zone. (b) The density contrast variation in 2-D spatial domain. (c) The binary 2-D SEG-EAGE salt model. (d) The resulting salt model with a depth-varying density contrast. (e) The simulated gravity data based on the density model (d) with Gaussian noise that has a standard deviation of 0.02 mGal.

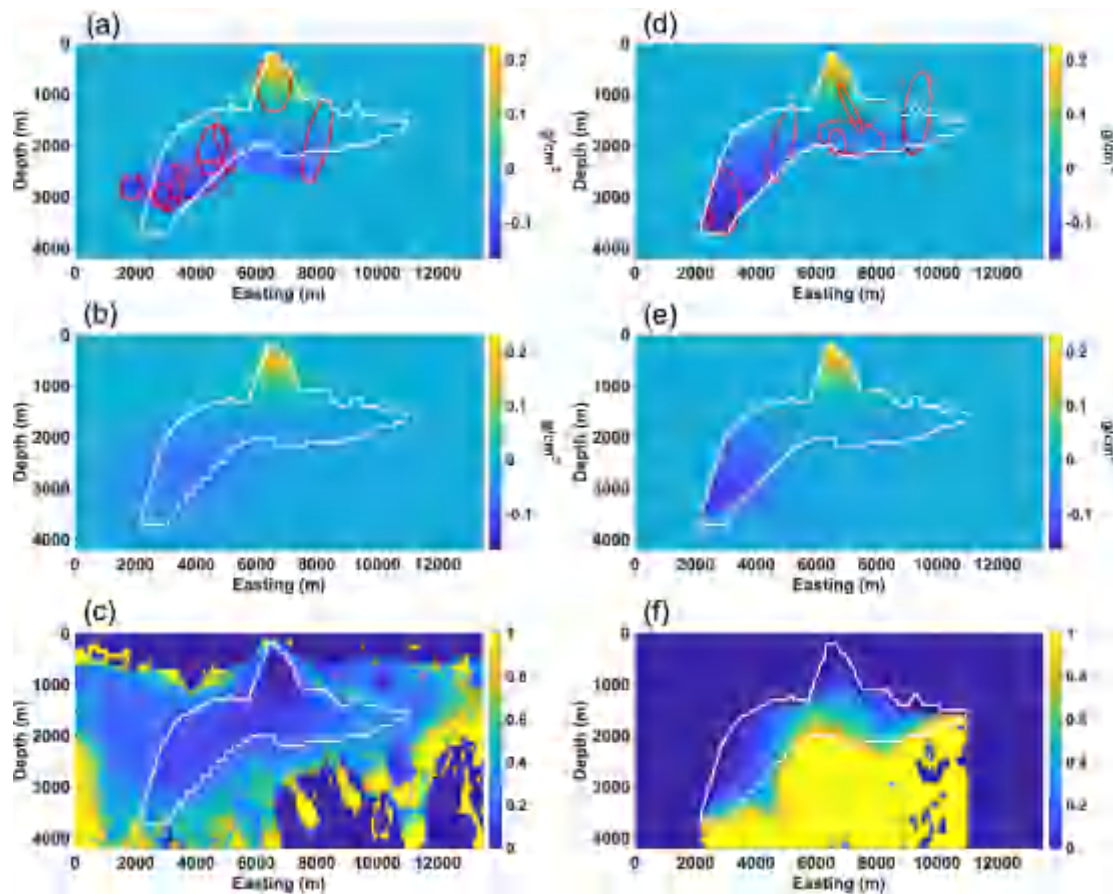


Figure 14. The recovered salt models with density variations in depth. Panels (a)–(c) indicate the recovered MAP model, reweighted mean and normalized standard deviations without any constraint (Fig. 5a). Panels (d)–(f) are the MAP model, reweighted mean and normalized standard deviations constrained by the deep informative constraint (Fig. 10a). The normalized standard deviations in (c) and (f) were obtained by normalizing the reweighted standard deviations (not shown) by the absolute values of the reweighted mean models, to better visualize the standard deviations around the nil zone.

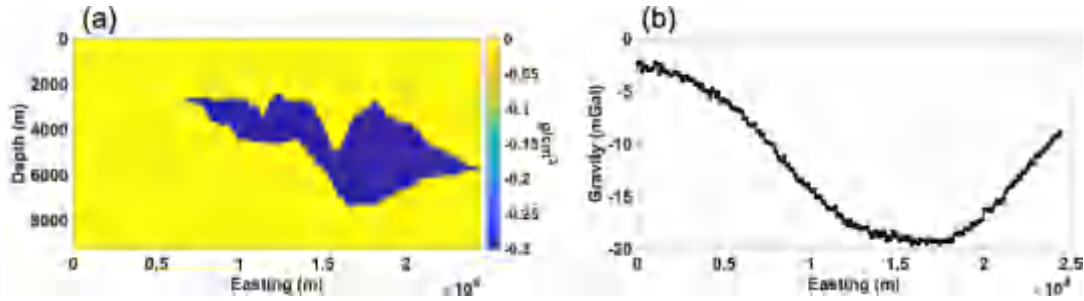


Figure 15. (a) The modified Sigsbee salt model with a constant salt density contrast value of -0.3 g cm^{-3} , which serves as the true model in our work. (b) The simulated gravity data based on the true model. The noise follows the Gaussian distribution with a standard deviation of 0.2 mGal.

4 Sigsbee SALT BODY RECONSTRUCTION

We apply our method to the modified Sigsbee 2B salt model that is characterized by a more complex geometry compared with the SEG-EAGE salt body. We firstly extract the salt body in the Sigsbee 2B synthetic model, and then assign the density contrast values of -0.3 and 0 g cm^{-3} to the salt and background, respectively. The modified model (Fig. 15a) consists a total of 26 700 cells (i.e. 267 by 100 in the east and depth). We simulated the observe gravity data and added the Gaussian noise with a standard deviation of 0.2 mGal (Fig. 15b). Because the two Sigsbee models (i.e. 2A and 2B) have the exact same salt geometry, hereafter, we do not distinguish them, and simply refer to the modified Sigsbee 2B salt model used in our work as the Sigsbee salt model.

In this section, again, we consider four different scenarios, namely, no constraint, conservative constraint, informative constraint as well as variation of density values with depth. In each scenario, we had 100 chains, each of which consisted of 20 000 steps. The sampling process took less than 4 hours using parallel computing. Similar to the SEG-EAGE salt model, we fixed 35 per cent of chains at the temperature level of 1 and randomly sampled the temperature levels from (1, 1500] for the reminder of the chains. We swapped samples among all the chains at each step based on the acceptance rule of randomized parallel tempering. We only kept the last 10 per cent samples with even indices in each chain for uncertainty analysis.

The parameters for the randomly sampled ellipses are summarized as follows. The number of ellipses was perturbed within [2, 14] following the *birth and death* strategy; the orientations were uniformly sampled from $[-\pi/2, \pi/2]$, and the semi-major and semi-minor axes were, respectively, drawn from [500 m, 2000 m] and [500 m, 1500m]. The centroid coordinates vary within the feasible region that is bounded by the red line shown in the Figs 16(a), (e) and (i). We also randomly sampled the tuning parameter (i.e. α) from [1000, 2000] to compute the concave hull shape.

4.1 Constant sedimentary background

The results for the three scenarios with a constant sedimentary background are summarized in Fig. 16. They are highly consistent with the observations for the SEG-EAGE model. In the scenario without using any constraint (Fig. 16a), the MAP model (Fig. 16b) and the reweighted mean model (Fig. 16c) reconstruct an overall average shape of the salt body at the correct depth, but the structural details of the salt body are poorly reconstructed. The areas occupied by negative density contrast materials (in blue in Fig. 16c) show relatively high standard deviations on the periphery and intermediate to low standard deviations in the interior, as shown in Fig. 16(d). The two tips, indicated by the red arrows, have near-zero standard deviations. However, similar to Fig. 5(d), the two tips are erroneously associated with sediments as shown in the mean model (Fig. 16c). Therefore, the near-zero standard deviations and uncertainties do not translate to high accuracy. In this case, our simulated gravity data do not contain necessary information that allows us to recover these two small tips, simply because their gravity response is exceedingly weak, and therefore, overwhelmed by noise in the data. The fact that our model is parametrized by ellipses, which are then connected through a concave hull, might add further to the difficulty in resolving these two small tips.

When deep conservative constraint is used (Fig. 16e), we observe improved recovery of the top boundary of the salt body (Fig. 16f). The spatial distribution of the salt materials is also more consistent with the shape of the salt body. In the reweighted mean density model (Fig. 16g), we notice a significant amount of negative density materials characterized by the deep blue colour and located at the centre of the anomalous body corresponding to near-zero standard deviations (Fig. 16h). Again, the two tips marked by red arrows should be interpreted with caution.

We also imposed the deep informative constraint (Fig. 16i) to improve the reconstruction of the Sigsbee salt model. As shown in Figs 16(j) and (k), both the MAP and reweighted mean models recover salt body satisfactorily. Fig. 16 l shows that the interior of the salt body is characterized by lower standard deviations, indicating that the accepted salt models agree on the existence of salt in these regions. Higher standard deviations roughly delineate the lower boundary of the salt body.

In addition, we constructed the Sigsbee salt probability models, following the same approach explained above for the SEG-EAGE salt model. Fig. 17(a) shows the salt probability model without any seismic constraint. we observe an area with high probabilities (of about 80 per cent) at roughly the average location of the true salt body. The overall shape is smooth, and the structural details of the salt body are lacking. We observe more structural details and higher probabilities when conservative (b) and informative (c) constraints are imposed.

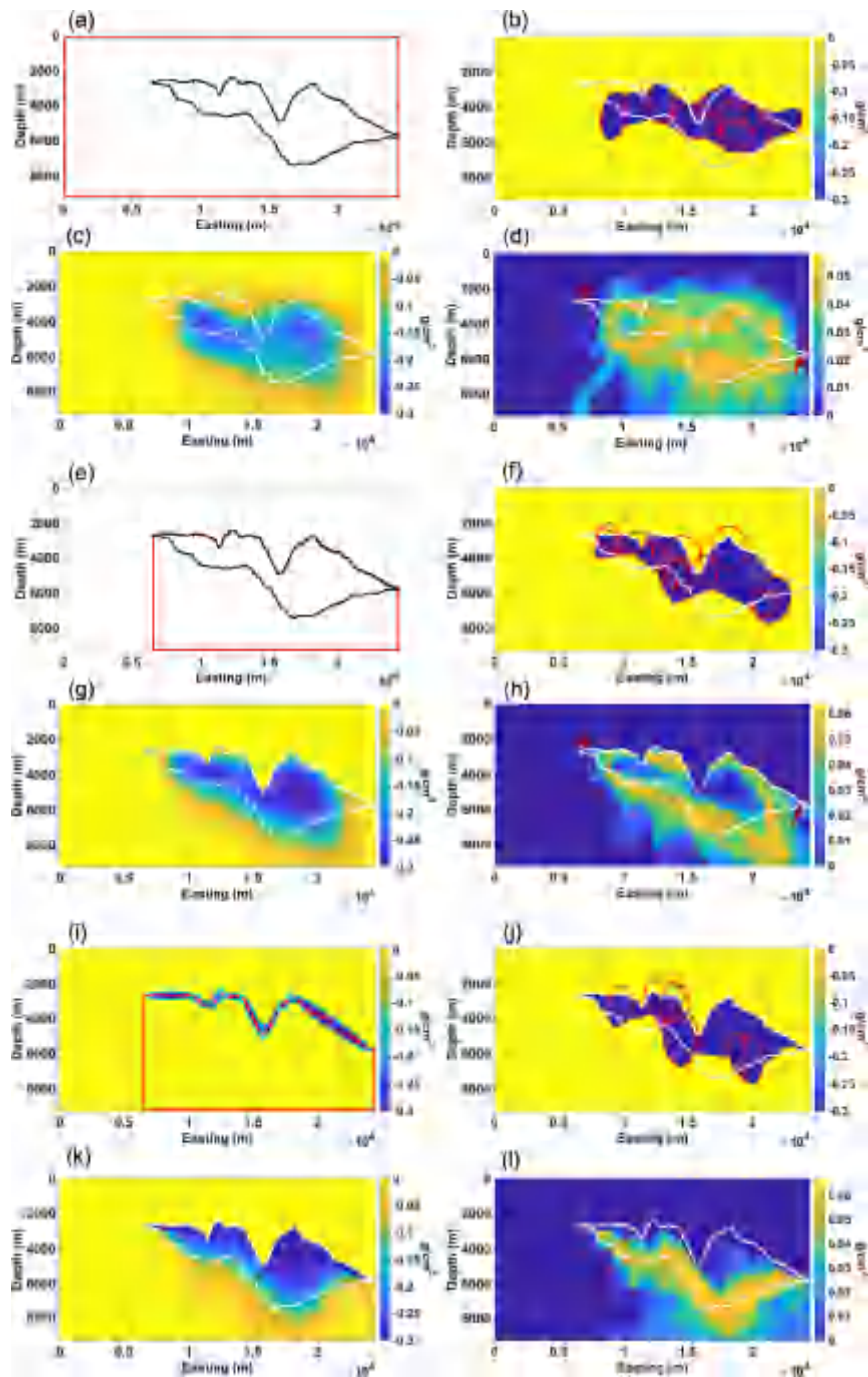


Figure 16. Panels (a)–(d) represent the scenario with no constraint. Panels (e)–(h) correspond to the deep conservative constraint. Panels (i)–(l) show the results using the deep informative constraint. The red and white lines have the same meaning as those shown in the figures for the SEG-EAGE salt model examples. All mean models and standard deviations are reweighted. Red arrows indicate the tips associated with lower standard deviations discussed in the text.

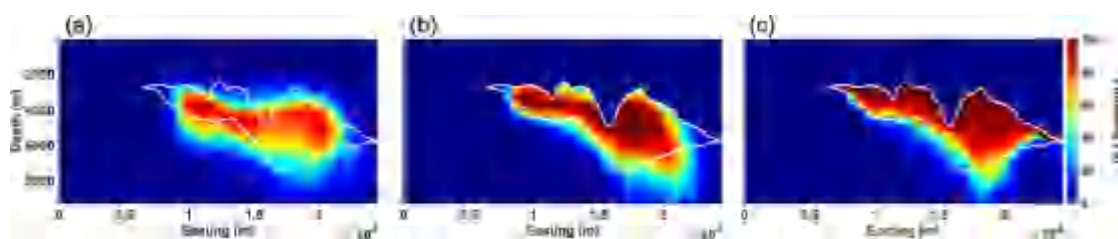


Figure 17. Probability of spatial distribution for the reconstructed Sigsbee salt models computed based upon the accepted models at the temperature level of 1. Panels (a) to (c) display the salt probability models corresponding to, respectively, no constraint on the top boundary (Fig. 16a), deep conservative constraint (Fig. 16e) and the deep informative constraint (Fig. 16i).

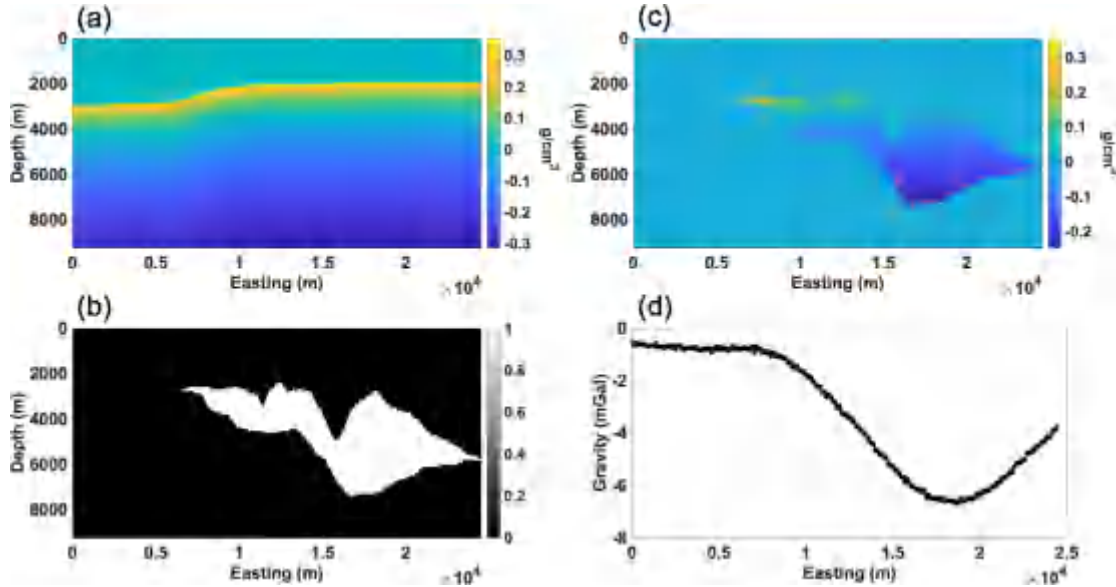


Figure 18. (a) The density contrast variation with depth following the relation in Fig. 13(a). The mudline is indicated by the yellow color associated with the highest density contrast value, above which is the sea water and below are sediments. (b) A binary representation of the modified Sigsbee salt model. (c) The Sigsbee salt model with density contrast variation in depth. (d) The simulated gravity data with Gaussian noise that has a standard deviation of 0.2 mGal.

4.2 Density contrast variation with depth

We use the same density versus depth relationship, that is $\rho = 1400 + 172 \times z^{0.21}$, for the Sigsbee model, where z is again the depth below the mudline. After removing the background density values in the seawater and the sedimentary rocks, we obtained a background density contrast model shown in Fig. 18(a). The warmest colour, that is yellow, represents the mudline provided with the Sigsbee model, above which is the sea water with an absolute density value of 1.02 g cm^{-3} and a density contrast of 0 g cm^{-3} . The binary Sigsbee salt model is displayed in Fig. 18(b). After the element-wise multiplication of the background density contrast model (Fig. 18a) and the binary model (Fig. 18b), we obtained the salt model with varied density contrast values (Fig. 18c), which serves as the true model in this section. In Fig. 18(c), we notice that the positive density contrast values concentrate at the shallower part of the salt body, whereas negative density contrast values distribute in the deeper part and occupy a much larger area. Based on the true model, we simulated the observed gravity data with a certain level of noise following a Gaussian distribution with a mean of 0 mGal and standard deviation of 0.05 mGal (Fig. 18d).

Next, we tried to reconstruct the shape of the true salt model (Fig. 18c) given the observed data (Fig. 18d). Similarly, we consider three different cases: no constraint, conservative constraint and informative constraint. Figs 19(a)–(c) corresponds to the case without using any constraint. The MAP model (Fig. 19a) shows two separately distributed anomalous bodies, namely, a small linear (or, tabular) body at the very top characterized by the positive density contrast and a sizeable body dominated by negative features. We observe several empty ellipses located near the surface, ranging from 0 to 10 000 m in easting and 0 to 3000 m in depth. These ellipses are located in the seawater, and therefore, were excluded from the calculation of the concave hull. The reweighted mean model (Fig. 19b) displays consistent features with the MAP model. We observe some positive and negative features, which correspond well to the lower standard deviations shown in Fig. 19(c). Overall, the recovered salt models in this case, albeit poor, are encouraging, considering that no constraint was used.

After imposing the conservative constraint (graphically represented by the feasible region in red in Fig. 16e), the reconstructed MAP model in Fig. 19(d) greatly improves compared with Fig. 19(a). The general shape of the salt body is reasonably recovered. Similar to the previous case, the positive and negative anomalous features displayed in the reweighted mean model (Fig. 19e) correspond to the low to near-zero standard deviations (Fig. 19f). In Fig. 19(f), we also notice that the lower standard deviations (blue colour except the background) match well with the overall shape of the true model and high standard deviations (yellow colour) roughly follow along the lower boundary of the salt body. Finally, when informative constraint (shown in Fig. 16i) was implemented, the MAP model (Fig. 19g) further improves, with the geometry of the salt body better defined and more consistent with the truth. The reweighted mean model shows a higher degree of agreement with the true model in Fig. 18(c). More encouragingly, the interior of the salt body is dominated by low to near-zero standard deviations. Similar to the previous cases, once again, we observe a reduction in the uncertainty when we move from no constraint (Fig. 19c), to conservative (Fig. 19f) and informative constraint (Fig. 19i).

5 DISCUSSIONS

The inverse problem of recovering salt bodies using only gravity measurements is ill-posed with many solutions and a great amount of uncertainties. Additional constraints from independent information based on other measurements must be imposed to obtain meaningful solutions. In our work, we focus on two types of prior information: the density contrast between salt and the sedimentary rocks and the top

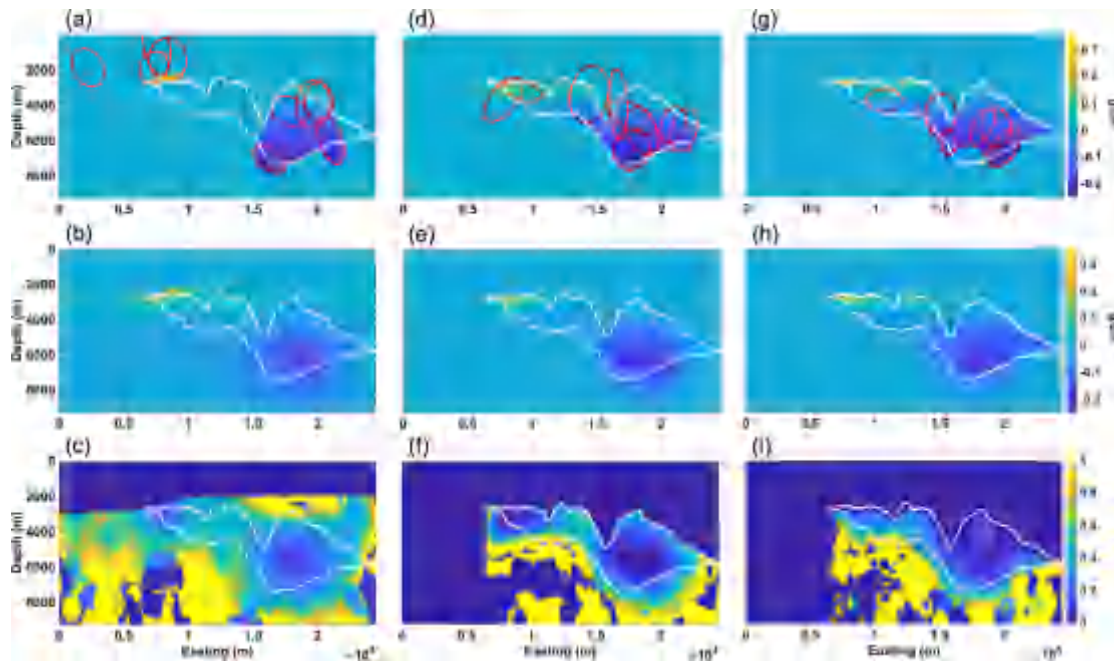


Figure 19. The reconstructed Sigsbee salt models when density contrast varies with depth. The recovered MAP (a), reweighted mean (b) and the normalized standard deviations (c) without using any constraint (Fig. 16a) are in the first column. The MAP, reweighted mean and standard deviations resulting from the deep conservative constraint (Fig. 16e) are displayed in (d), (e) and (f), respectively. (g), (h) and (i) show, respectively, the MAP and reweighted mean and standard deviations resulting from the deep informative constraint (Fig. 16i). We note that the normalized standard deviations are displayed here to better visualize the standard deviations around the zero zone where the density contrast is 0 g cm^{-3} .

boundary of the salt body imaged by seismic data. Our work shows quantitatively the increase in the uncertainties when the density contrast value becomes more uncertain (Fig. 12). Several different constraints on the top of the salt body from seismic imaging were implemented by designing feasible regions and placing fixed ellipses. Overall, our work shows that these additional constraints greatly promote the salt body recovery and reduce uncertainties. Other types of measurements can also be used to further constrain the recovery of the salt body shapes. For example, borehole gravity measurements can provide depth information and help better recover the bottom boundary of the salt body. The gravity gradient measurements are more sensitive to the lateral changes in the geometry of the salt body. We believe that the methodology that we have developed can be easily extended to integrate multiple geophysical data sets in imaging salt bodies.

Monte Carlo sampling is computationally prohibitive due to the curse of dimensionality. We use the sparse geometry parametrization to reduce the model dimensions (i.e. the number of parameters to be sampled). Specifically, we use a set of simple geometries, each of which is parametrized by only a few parameters. The 2D SEG/EAGE and the Sigsbee salt body shapes were adequately approximated by less than 20 ellipses, which translates to MCMC sampling in a much lower dimensional space (less than 100) than would otherwise have required if a conventional cell-based parametrization was used. The low-dimensional Monte Carlo sampling helps us address the computational challenges. In our work, two million SEG-EAGE models were sampled within 2 hr and the Sigsbee models took less than 4 hr. Given the computational efficiency provided by the sparse geometry parametrization, a logical next step would be to extend it to 3-D Bayesian inference problems and 3-D joint inversions.

Luo (2010) and Titus *et al.* (2017) recovered the geometry of an anomalous body in the MCMC framework. The geometry is parametrized by the coordinates of a polygon's vertices. While polygons also provide sparse parametrization of complex geometries, extra care needs to be taken to make sure that the polygons have reasonable shapes. Luo (2010) and Titus *et al.* (2017) use, respectively, angles and perimeters to constrain the shapes of the sampled polygons. In our work, such quality control measures are not necessary thanks to the computation of concave hull. In addition, the use of parallel tempering and investigation of different constraints also distinguish our work from theirs.

Our work clearly shows that low uncertainty does not necessarily translate to high accuracy, as shown in Figs 5(d) and 16 and discussed in the relevant text. The low uncertainty might result from the lack of information in the data that is necessary for resolving the features of interest. In the case of the Sigsbee model, the gravity response of the two tips at the western and eastern edges is simply too weak to allow for reliable estimates of the salt materials at these two locations, unless independent and strong constraint is imposed. This is not a failure or weakness of the MCMC algorithm, but rather an inherent feature of it. Practitioner should exercise care when interpreting the mean and standard deviation results by considering the amount of prior information used in the sampling algorithms, the resolution, resolving power and uncertainty of geophysical data as well as the chosen parametrization strategy.

We emphasize that there is no one, single 'true' model uncertainty that an uncertainty quantification algorithm is supposed to find. The uncertainties resulting from a certain algorithm have to do with prior information or constraint imposed on the inverse problem. When prior constraints change, the estimates of uncertainties would also change, as evidenced by the results from different scenarios considered in our

work. Any uncertainty quantification is only as good as the prior information supplied to the algorithm that produced it. The uncertainties are only meaningful and ‘correct’ when interpreted properly in light of the prior information, data uncertainty and data resolution. Unfortunately, current practices of interpreting uncertainty quantification results are oversimplified, and often, only focus on the relative magnitudes of the resulting standard deviation values. We see a significant and urgent need to develop more insights and tools for the proper interpretation of uncertainty quantification results.

6 CONCLUSIONS

Quantifying the uncertainties of the salt body shapes provides important information for decision makers when evaluating risks associated with drilling. It also allows us to understand the added value of additional constraints from, say, seismic imaging. We estimate the salt body shapes and quantify their uncertainties using gravity data in the trans-dimensional Bayesian framework. To tackle the computational challenges associated with MCMC sampling, we use the sparse geometry parametrization, which allows the complex shapes of salt bodies to be approximated by a set of simple geometries such as ellipses, while greatly reducing the number of parameters to be sampled. We also implement the randomized parallel tempering method to improve the acceptance ratios during sampling.

To understand how the uncertainties of the recovered salt body shapes change with different constraints, we design five different scenarios. When only gravity measurements are used without any seismic constraint, we can only recover an overall average shape of the salt body with the structural details lacking. When the top of the salt bodies are constrained by seismic imaging, the recovered salt bodies show much improvement with better defined geometries. The interior of the recovered salt bodies is also characterized by larger areas of lower uncertainties. To further constrain the salt body recovery, we propose a new method of using both fixed and random ellipses that are connected by their concave hull. We find that, by constraining the salt materials immediately beneath the top boundary of salt bodies using this new method, the salt body shapes are further improved and the areas of low uncertainties within salt bodies are increased.

Our numerical results based on modified SEG-EAGE and Sigsbee salt models clearly and quantitatively shows that, the uncertainties of the recovered salt bodies from gravity measurements are reduced when the degree of the seismic constraint on the top boundary of salt bodies becomes stronger and the uncertainty of the density contrast values decreases. Although this finding has been well recognized, the quantification of the uncertainty reduction and systematic comparison in different scenarios in a rigorous Bayesian inference framework is an important and novel contribution of our work.

To make our uncertainty analysis results more relevant for the geophysical practitioners, we compute the salt probabilities models for all the scenarios considered in our work. These models provide an intuitive picture of the spatial distribution of salt and its probabilities, and are arguably more useful than the most widely visualized mean, MAP and standard deviation models from a practical point of view.

Our work indicates that low uncertainty does not necessarily translate to high accuracy. Low uncertainty can occur in both well-resolvable and poorly resolvable areas in a model. For the former, low uncertainty typically corresponds to high accuracy, whereas for the latter, it does not. Uncertainties are only meaningful and ‘correct’ when interpreted properly in light of the prior information, data uncertainty, data resolution and model parametrization.

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DATA AVAILABILITY

The gravity data used in this study are all synthetic based on the modified SEG/EAGE and Sigsbee salt models. They are available upon request to the corresponding author.

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APPENDIX A

A1 The acceptance rule for model exchange in tempering MCMC

Parallel tempering consists of a swap of models at two temperature levels and is formalized as following (Sambridge, 2014):

$$(\mathbf{m}^i, T_i), (\mathbf{m}^j, T_j) \rightarrow (\mathbf{m}^i, T_j), (\mathbf{m}^j, T_i), \quad (\text{A1})$$

where the left-hand side of eq. (A1) contains two sampled models at the current stage, $\mathbf{m}^i$ and $\mathbf{m}^j$, at two different temperature levels, T_i and T_j . The right-hand side represents a candidate stage where the two models are swapped. The parallel tempering provides a way to sample the combined distribution. For instance, the joint posterior probability for the left-hand side of the eq. (A1) can be written as:

$$\frac{P(\mathbf{m}^i | \mathbf{d}^{\text{obs}})^{1/T_i} Q(T_j | T_i)}{C(T_i)} \times \frac{P(\mathbf{m}^j | \mathbf{d}^{\text{obs}})^{1/T_j} Q(T_i | T_j)}{C(T_j)}, \quad (\text{A2})$$

where $P(\cdot)$, $Q(\cdot)$ and $C(\cdot)$ represent the posterior distribution, proposal distribution and normalizing constants, respectively.

According to the Metropolis–Hastings rule, the acceptance ratio for the model exchange can be formalized as:

$$\alpha = \min \left(1, \frac{P(\mathbf{m}^i | \mathbf{d}^{\text{obs}})^{1/T_j} Q(T_i | T_j) C(T_i) \times P(\mathbf{m}^j | \mathbf{d}^{\text{obs}})^{1/T_i} Q(T_j | T_i) C(T_j)}{P(\mathbf{m}^j | \mathbf{d}^{\text{obs}})^{1/T_i} Q(T_j | T_i) C(T_j) \times P(\mathbf{m}^i | \mathbf{d}^{\text{obs}})^{1/T_j} Q(T_i | T_j) C(T_i)} \right), \quad (\text{A3})$$

where $Q(\cdot)$ and $C(\cdot)$ can be cancelled out. After simplification, we obtain:

$$\alpha = \min \left(1, \left[\frac{P(\mathbf{m}^i | \mathbf{d}^{\text{obs}})}{P(\mathbf{m}^j | \mathbf{d}^{\text{obs}})} \right]^{1/T_j} \times \left[\frac{P(\mathbf{m}^j | \mathbf{d}^{\text{obs}})}{P(\mathbf{m}^i | \mathbf{d}^{\text{obs}})} \right]^{1/T_i} \right), \quad (\text{A4})$$

Equivalently, we obtain:

$$\alpha = \min \left(1, \exp \left[\left(\frac{1}{T_j} - \frac{1}{T_i} \right) (\phi(\mathbf{m}^j) - \phi(\mathbf{m}^i)) \right] \right), \quad (\text{A5})$$

where $\phi(\cdot)$ represents the objective function in the deterministic framework.

APPENDIX B

B1 An importance weight in tempering MCMC

A large set of samples drawn from a distribution can help approximate integrals. For example, the mean and variance can be generically written in the form of integrals over the PPD and a function. The goal of the reweighting procedure in tempering MCMC is to compute unbiased integral estimates (i.e. mean and variance) with $T = 1$ using the samples collected from a tempered chain T_k ($T_k > 1$).

$$\bar{\mathbf{m}} = \int_{\mathbf{M}} \mathbf{m} P(\mathbf{m} | \mathbf{d}^{\text{obs}})^{1/T_1} d\mathbf{m} = \int_{\mathbf{M}} \mathbf{m} \frac{P(\mathbf{m} | \mathbf{d}^{\text{obs}})^{1/T_1}}{P(\mathbf{m} | \mathbf{d}^{\text{obs}})^{1/T_k}} P(\mathbf{m} | \mathbf{d}^{\text{obs}})^{1/T_k} d\mathbf{m}, \quad (\text{B1})$$

Eq. (B1) shows that the mean of the models sampled from the PPD at T_1 can be approximated by a PPD at T_k scaled by the ratio of $\frac{P(\mathbf{m} | \mathbf{d}^{\text{obs}})^{1/T_1}}{P(\mathbf{m} | \mathbf{d}^{\text{obs}})^{1/T_k}}$. We note that eq. (B1) is always true for the models $\mathbf{m}$ sampled from both T_1 and T_k .

According to the Bayes' rule, the right-hand side of the eq. (B1) can be formalized as:

$$\bar{\mathbf{m}} = \int_{\mathbf{M}} \mathbf{m} \frac{L(\mathbf{d}^{\text{obs}} | \mathbf{m})^{1/T_1} C_{T_k}}{L(\mathbf{d}^{\text{obs}} | \mathbf{m})^{1/T_k} C_{T_1}} \frac{L(\mathbf{d}^{\text{obs}} | \mathbf{m})^{1/T_k} P(\mathbf{m})}{C_{T_k}} d\mathbf{m}, \quad (\text{B2})$$

where C_{T_k} and C_{T_1} are the normalization constants for the chains associated with T_k and T_1 . $L(\cdot)$ and $P(\cdot)$ represent the likelihood and prior distributions, respectively. Rearranging eq. (B2), we obtain:

$$\bar{\mathbf{m}} = \frac{C_{T_k}}{C_{T_1}} \int_{\mathbf{M}} \mathbf{m} L(\mathbf{d}^{\text{obs}} | \mathbf{m})^{(1/T_1 - 1/T_k)} \frac{L(\mathbf{d}^{\text{obs}} | \mathbf{m})^{1/T_k} P(\mathbf{m})}{C_{T_k}} d\mathbf{m}, \quad (\text{B3})$$

$$\bar{\mathbf{m}} \approx \frac{C_{T_k}}{C_{T_1}} \frac{1}{N} \sum_{j=1}^N \mathbf{m}_k^j L(\mathbf{d}^{\text{obs}} | \mathbf{m}_k^j)^{(1/T_1 - 1/T_k)}, \quad (\text{B4})$$

where $\mathbf{m}_k^j$ is the j th model sampled from the chain tempered by T_k with a total steps of N .

Given $C_{T_1} = \int_{\mathbf{M}} L(\mathbf{d}^{\text{obs}}|\mathbf{m})^{1/T_1} P(\mathbf{m}) d\mathbf{m}$, divided by C_{T_k} in the both sides, we obtain:

$$\frac{C_{T_1}}{C_{T_k}} = \int_{\mathbf{M}} L(\mathbf{d}^{\text{obs}}|\mathbf{m})^{(1/T_1 - 1/T_k)} \frac{L(\mathbf{d}^{\text{obs}}|\mathbf{m})^{1/T_k} P(\mathbf{m})}{C_{T_k}} d\mathbf{m}, \quad (\text{B5})$$

$$\frac{C_{T_1}}{C_{T_k}} \approx \frac{1}{N} \sum_{j=1}^N L(\mathbf{d}^{\text{obs}}|\mathbf{m}_k^j)^{(1/T_1 - 1/T_k)}, \quad (\text{B6})$$

Combining eqs (B4) and (B6), the unbiased posterior mean model can be written as:

$$\bar{\mathbf{m}} \approx \frac{\sum_{j=1}^N \mathbf{m}_k^j L(\mathbf{d}^{\text{obs}}|\mathbf{m}_k^j)^{(1/T_1 - 1/T_k)}}{\sum_{j=1}^N L(\mathbf{d}^{\text{obs}}|\mathbf{m}_k^j)^{(1/T_1 - 1/T_k)}}, \quad (\text{B7})$$

Defining $\gamma_k^j = L(\mathbf{d}^{\text{obs}}|\mathbf{m}_k^j)^{(1/T_1 - 1/T_k)}$, eq. (B7) can be simplified as follows:

$$\bar{\mathbf{m}} \approx \frac{\sum_{j=1}^N \gamma_k^j \mathbf{m}_k^j}{\sum_{j=1}^N \gamma_k^j}, \quad (\text{B8})$$

The unbiased posterior variance is:

$$\text{var} \approx \frac{\sum_{j=1}^N \gamma_k^j (\mathbf{m}_k^j - \bar{\mathbf{m}})^2}{\sum_{j=1}^N \gamma_k^j}, \quad (\text{B9})$$

Eqs. (B8) and (B9) show the unbiased mean and variance for one chain at the temperature level of T_k . Expanding these two equations to a sequence of chains (a total of C chains) with different temperatures, $T \in [T_1, T_2, \dots, T_k, \dots, T_c]$.

The unbiased mean model and variance for all chains are defined as:

$$\bar{\mathbf{m}} \approx \frac{1}{C} \sum_{k=1}^C \frac{\sum_{j=1}^N \gamma_k^j \mathbf{m}_k^j}{\sum_{j=1}^N \gamma_k^j}, \quad (\text{B10})$$

$$\text{var} \approx \frac{1}{C} \sum_{k=1}^C \frac{\sum_{j=1}^N \gamma_k^j (\mathbf{m}_k^j - \bar{\mathbf{m}})^2}{\sum_{j=1}^N \gamma_k^j}. \quad (\text{B11})$$